

BFS Traversal

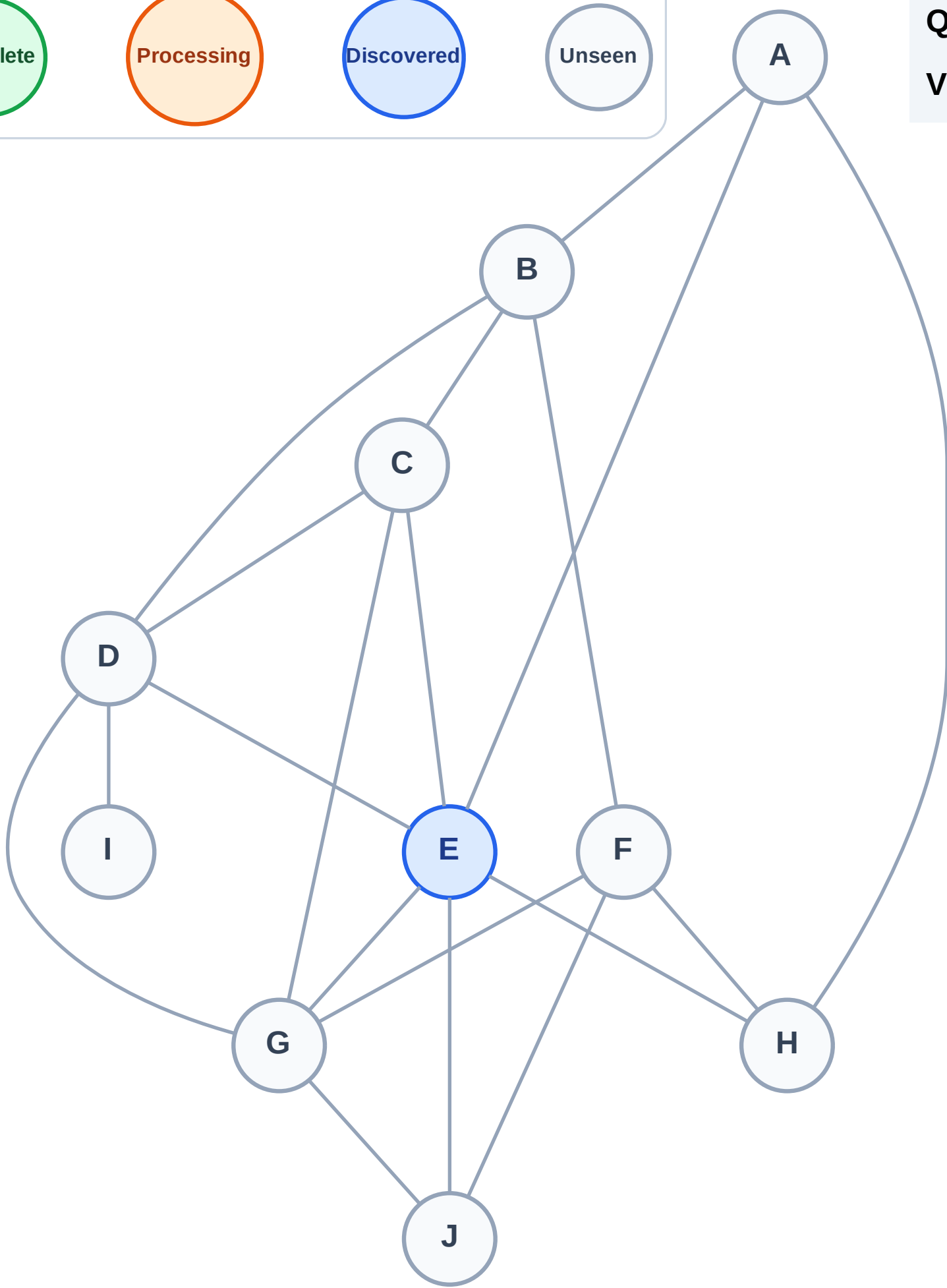
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

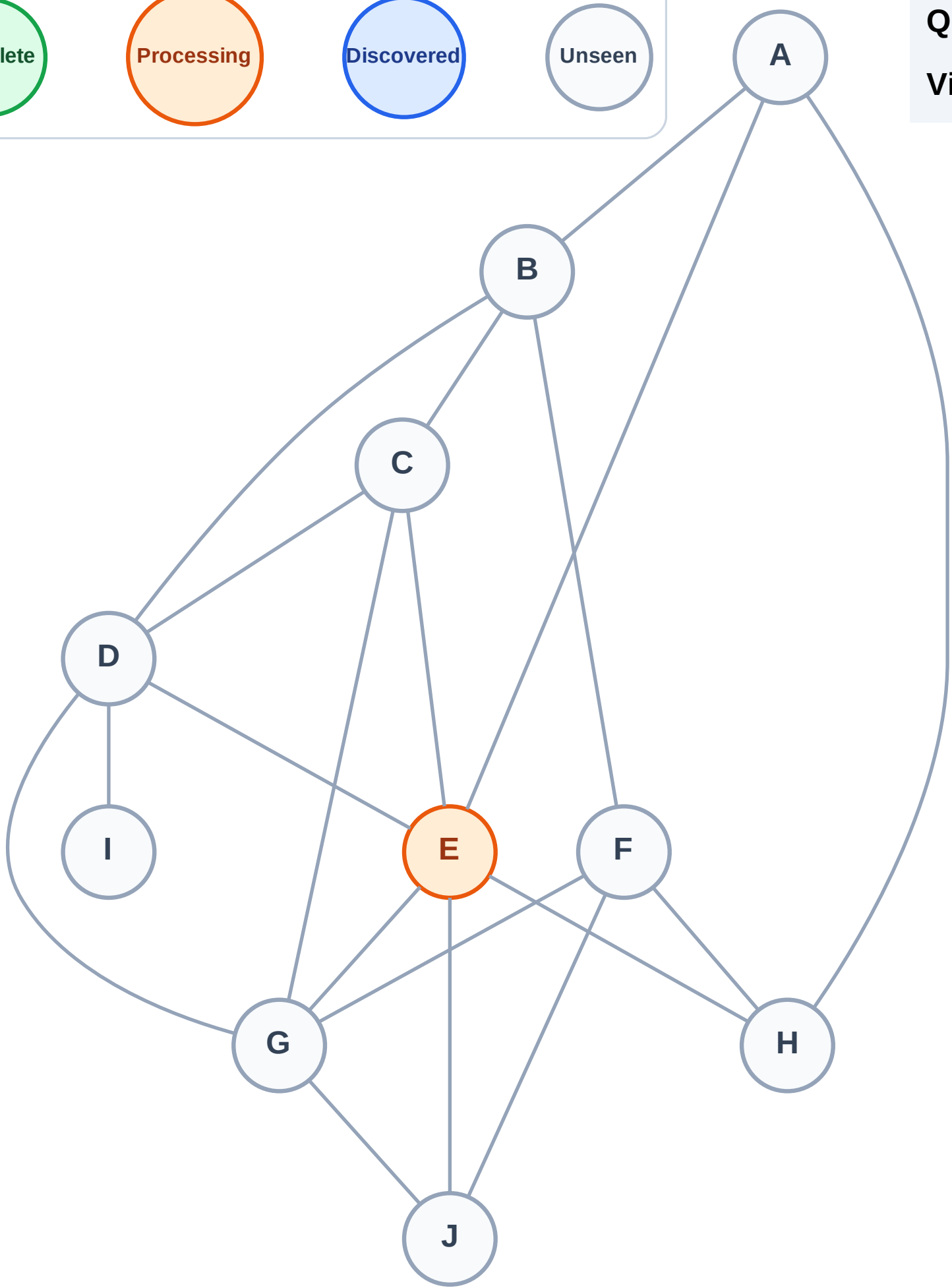
Step 2: Process node E

Legend



Queue: (empty)

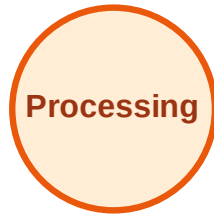
Visited: (none)



BFS Traversal

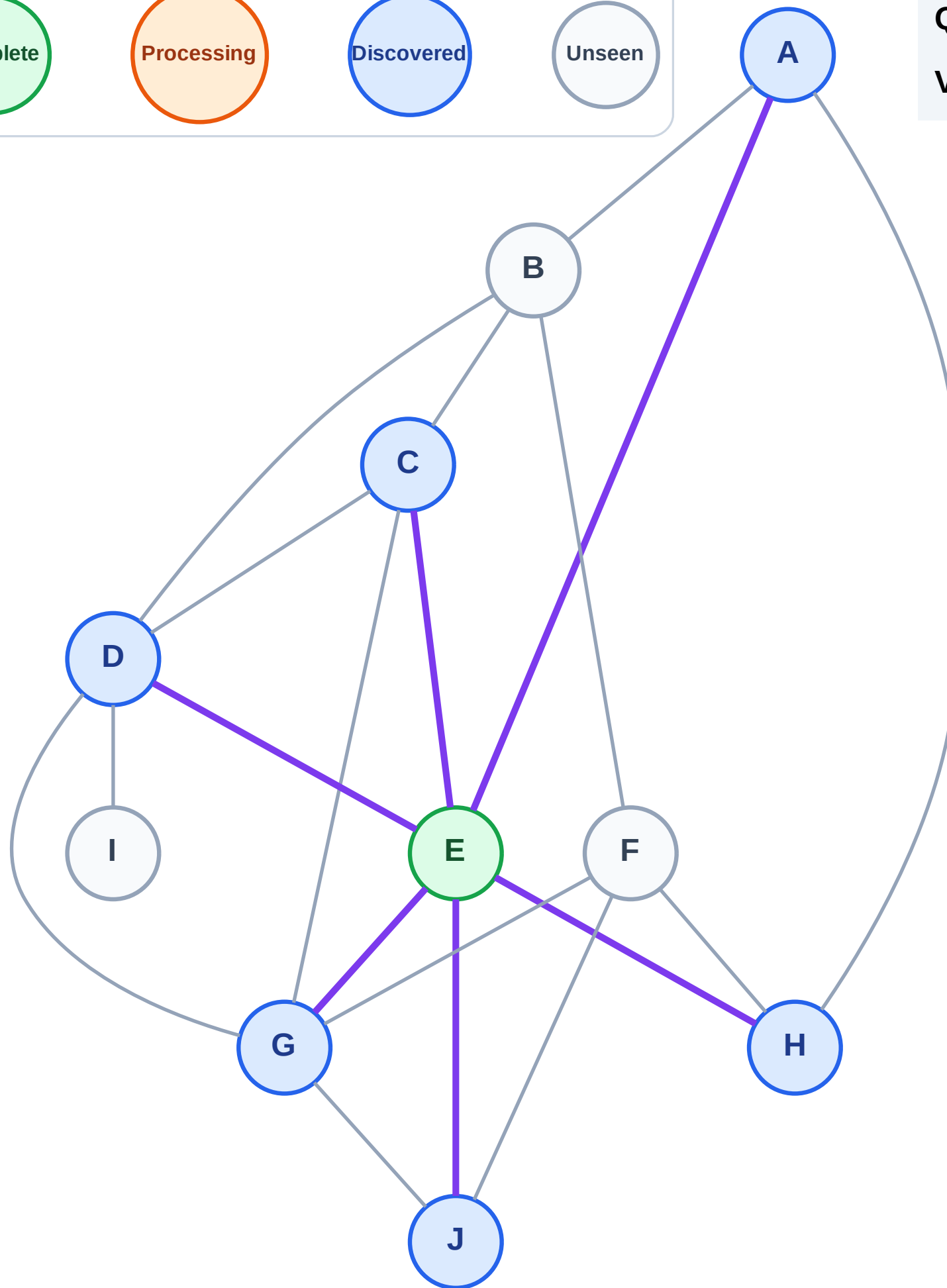
Step 3: Discover A, C, D, G, H, J from E

Legend



Queue: A → C → D → G → H → J

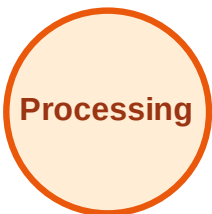
Visited: E



BFS Traversal

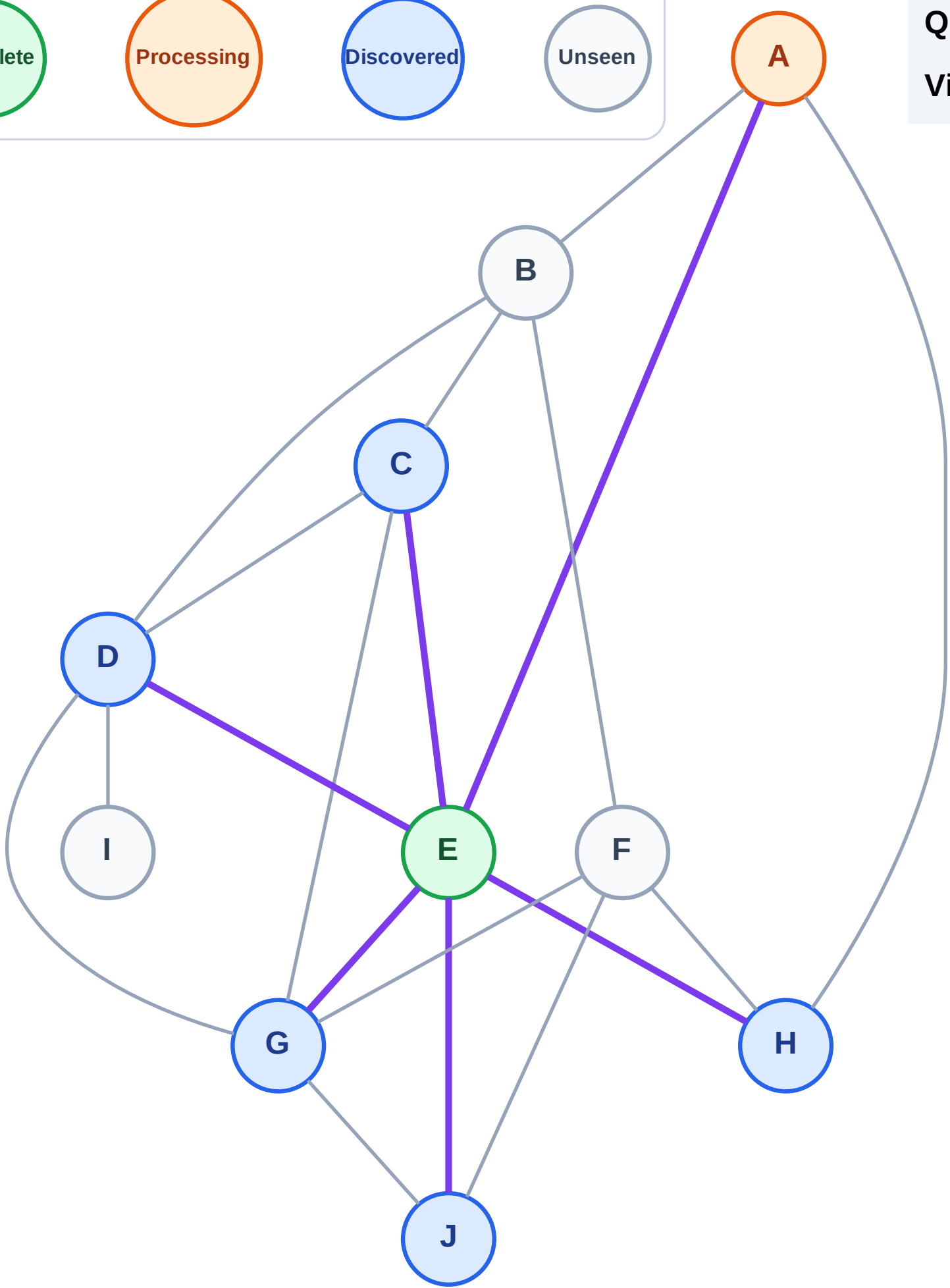
Step 4: Process node A

Legend



Queue: C → D → G → H → J

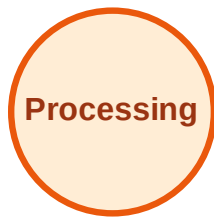
Visited: E



BFS Traversal

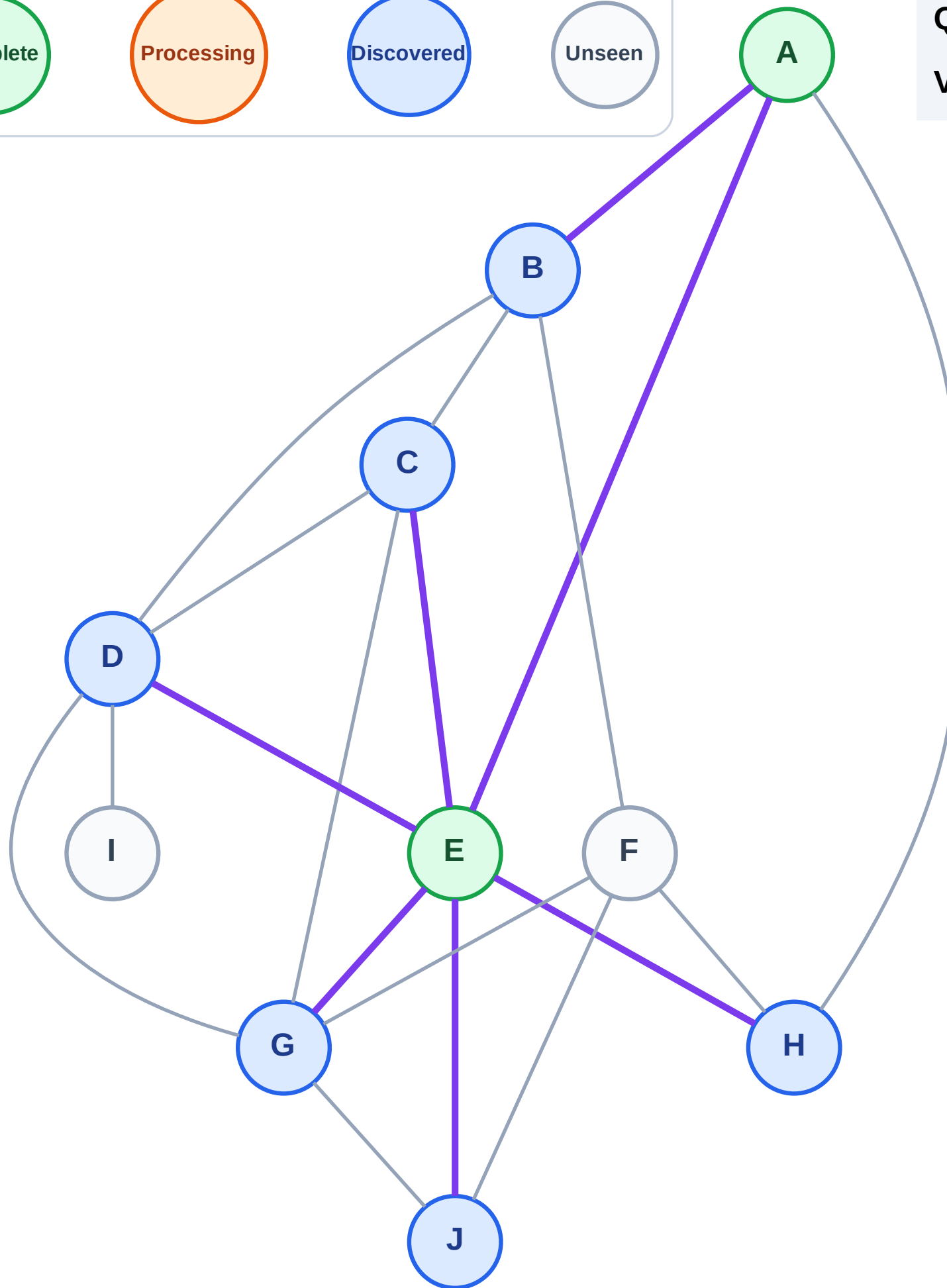
Step 5: Discover B from A

Legend



Queue: C → D → G → H → J → B

Visited: A, E



BFS Traversal

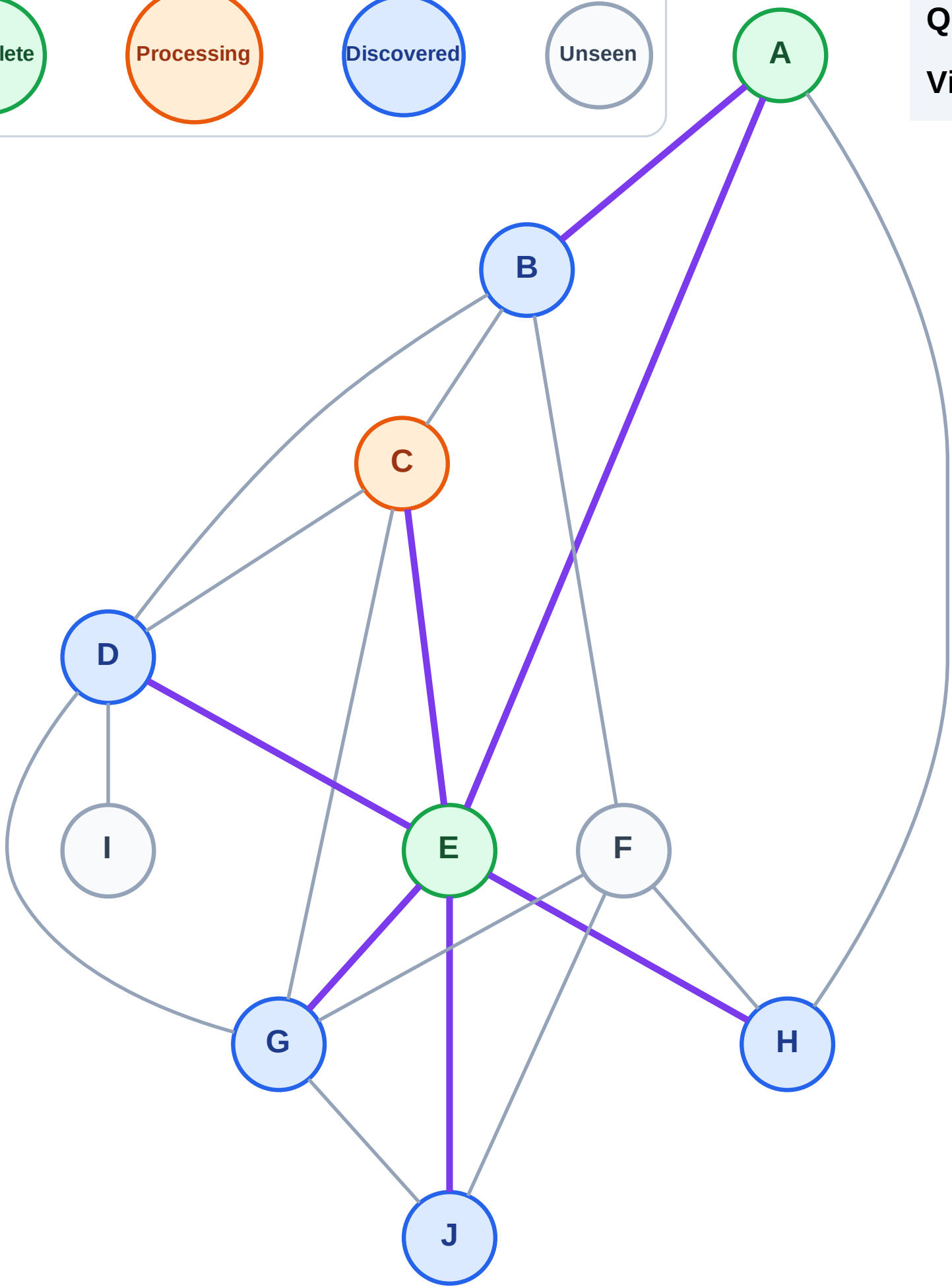
Step 6: Process node C

Legend



Queue: D → G → H → J → B

Visited: A, E



BFS Traversal

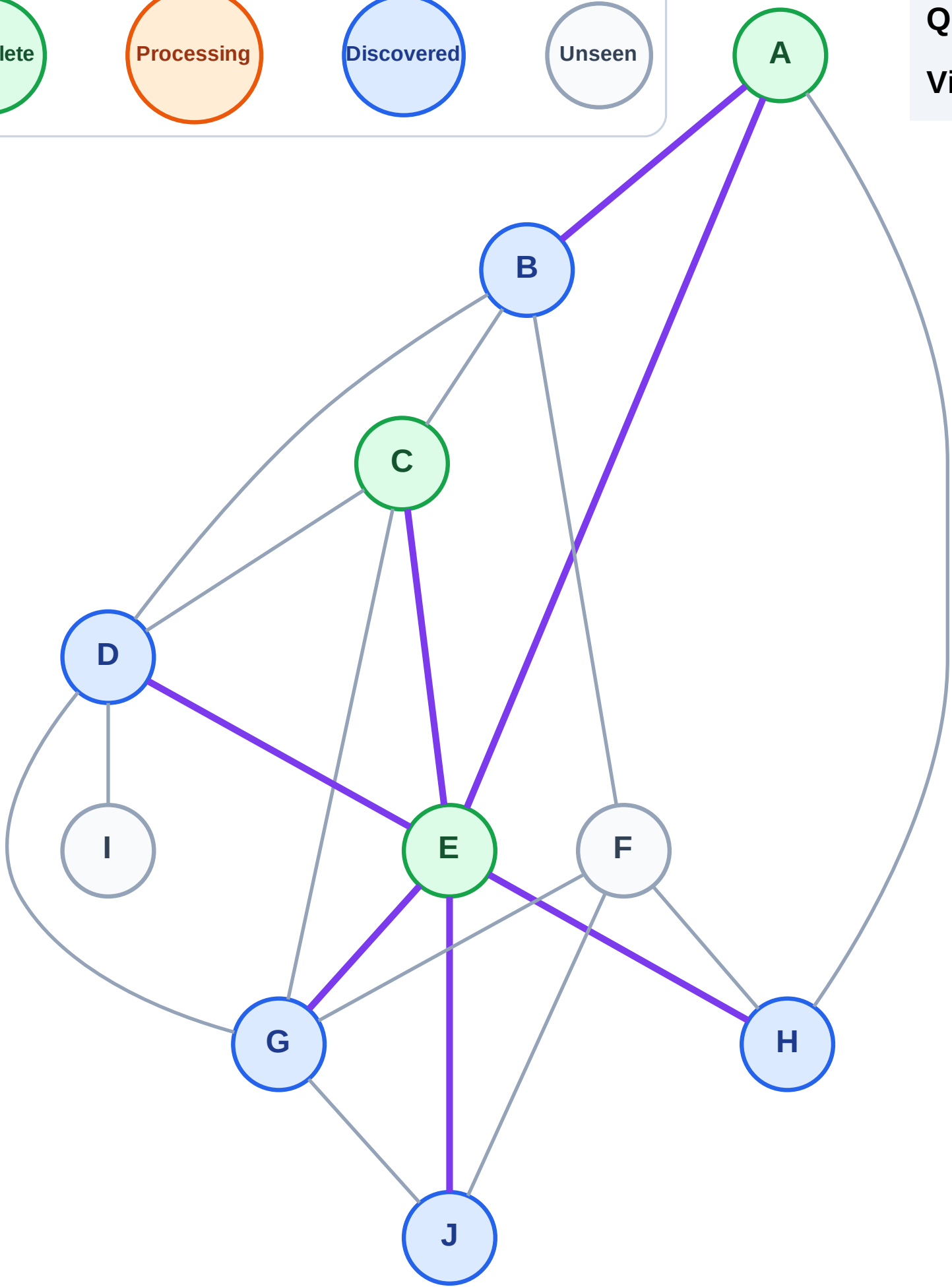
Step 7: No new neighbors from C

Legend



Queue: D → G → H → J → B

Visited: A, C, E



BFS Traversal

Step 8: Process node D

Legend

Complete

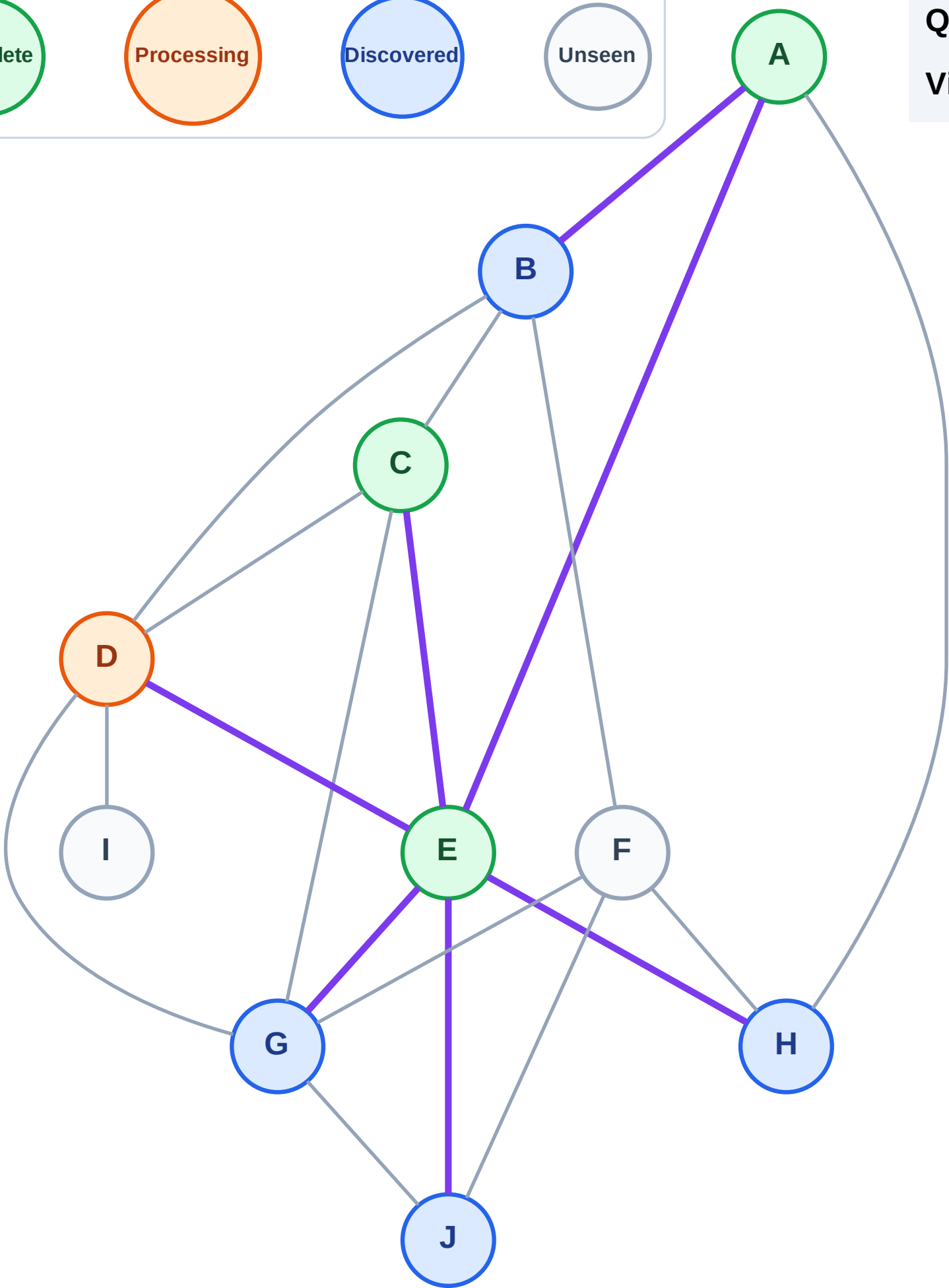
Processing

Discovered

Unseen

Queue: G → H → J → B

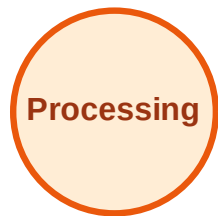
Visited: A, C, E



BFS Traversal

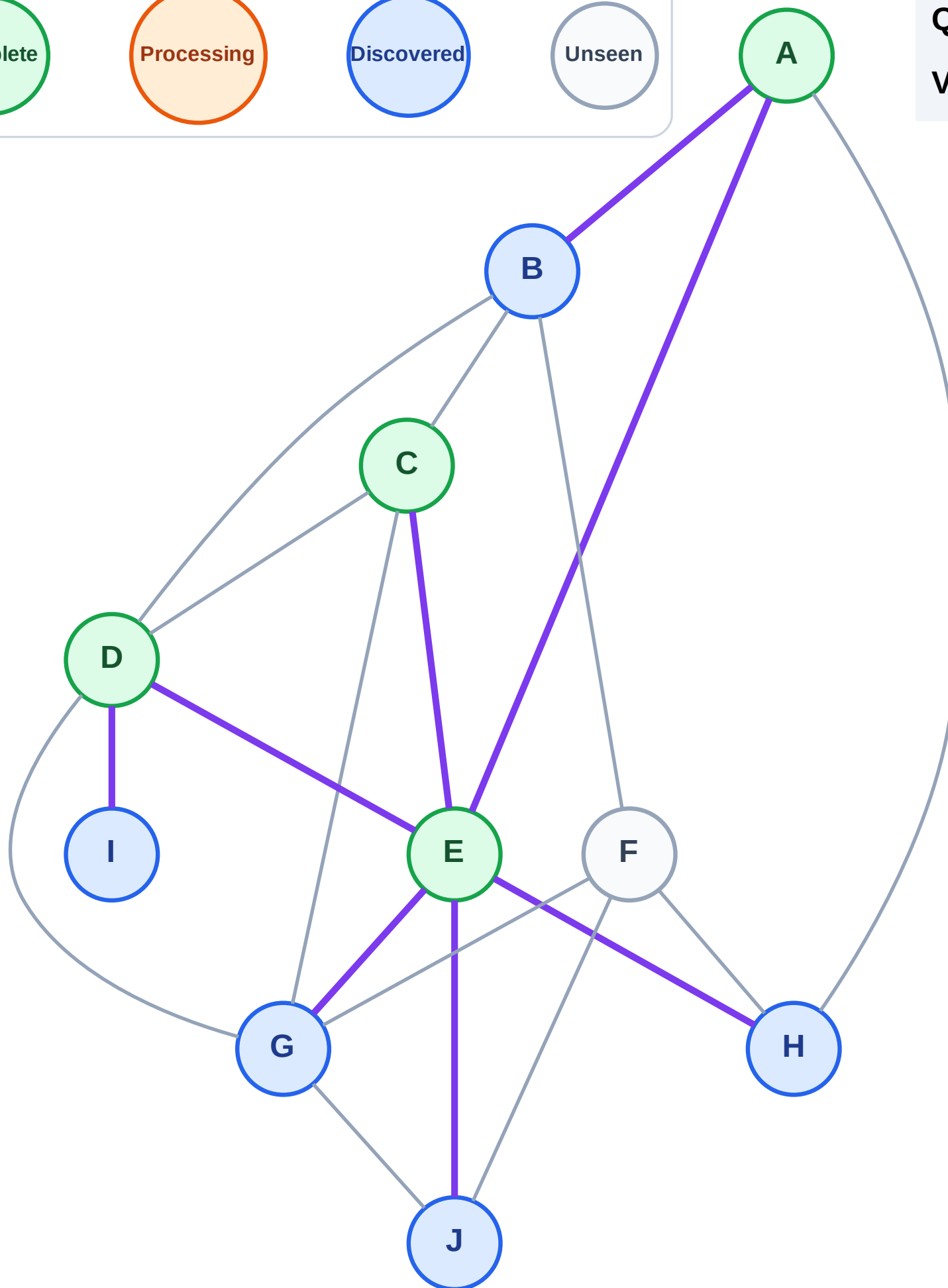
Step 9: Discover I from D

Legend



Queue: G → H → J → B → I

Visited: A, C, D, E



BFS Traversal

Step 10: Process node G

Legend

Complete

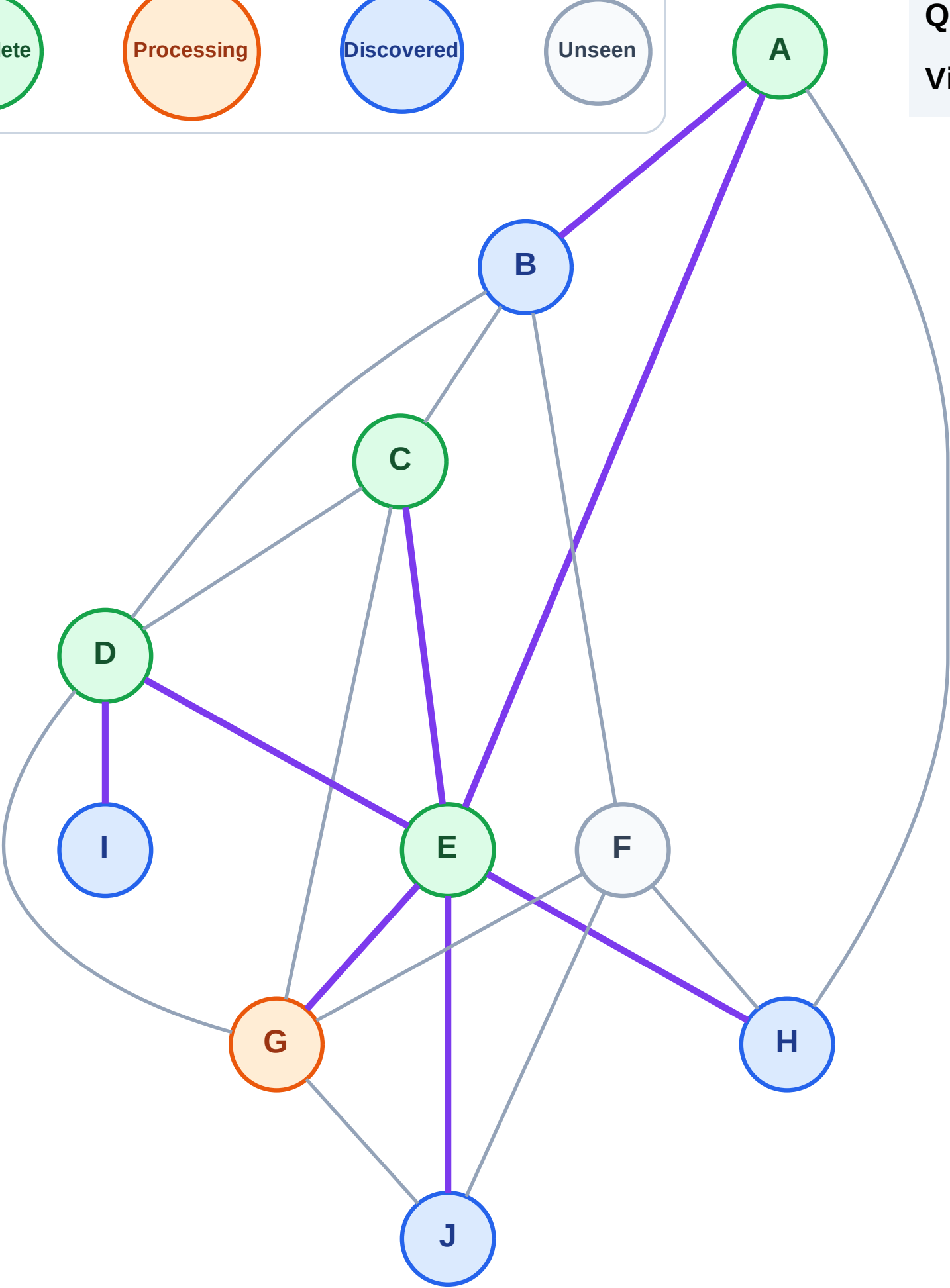
Processing

Discovered

Unseen

Queue: H → J → B → I

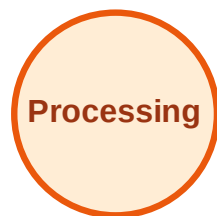
Visited: A, C, D, E



BFS Traversal

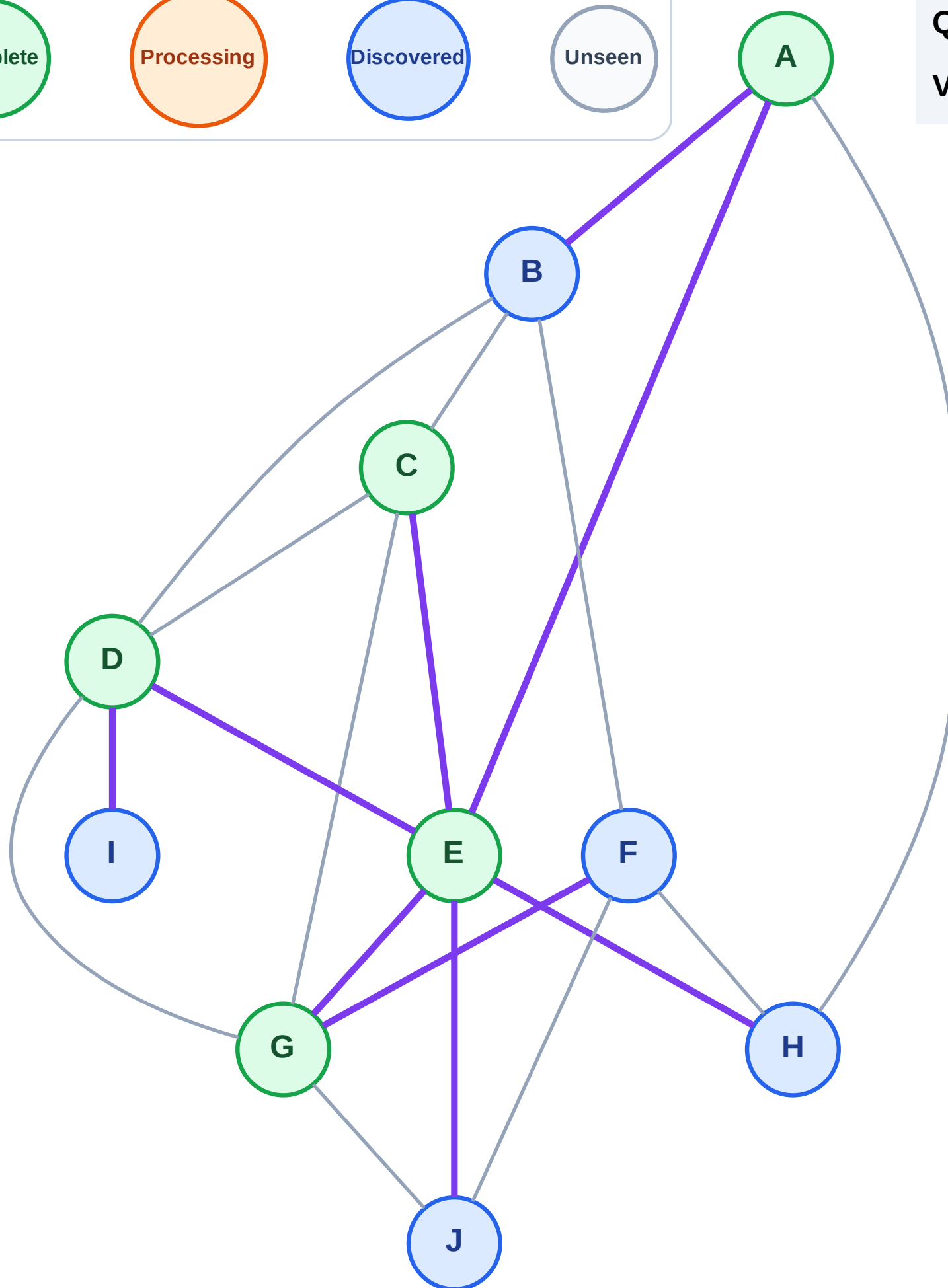
Step 11: Discover F from G

Legend



Queue: H → J → B → I → F

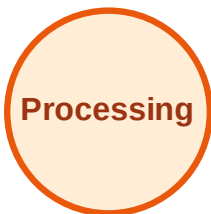
Visited: A, C, D, E, G



BFS Traversal

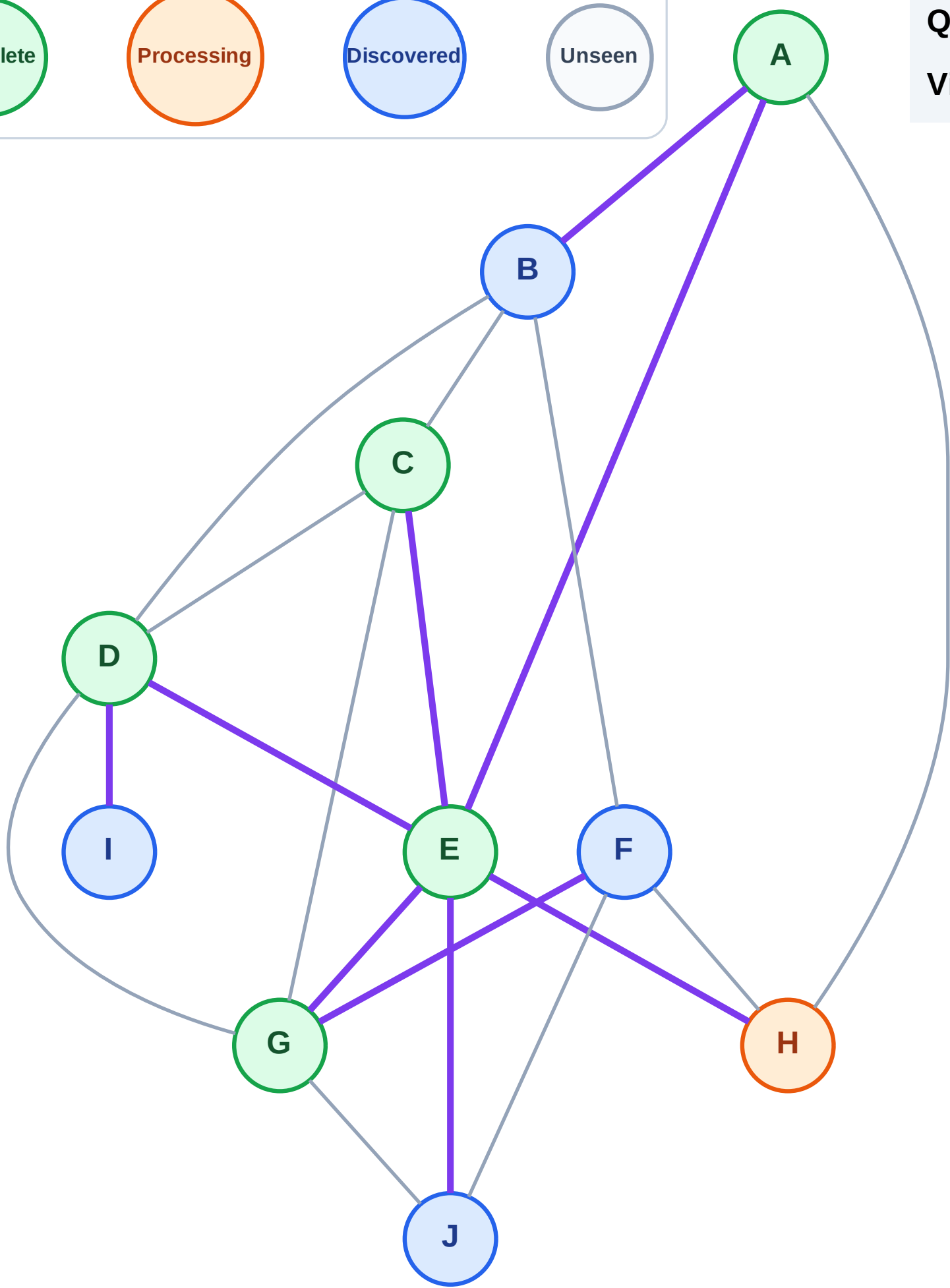
Step 12: Process node H

Legend



Queue: J → B → I → F

Visited: A, C, D, E, G



BFS Traversal

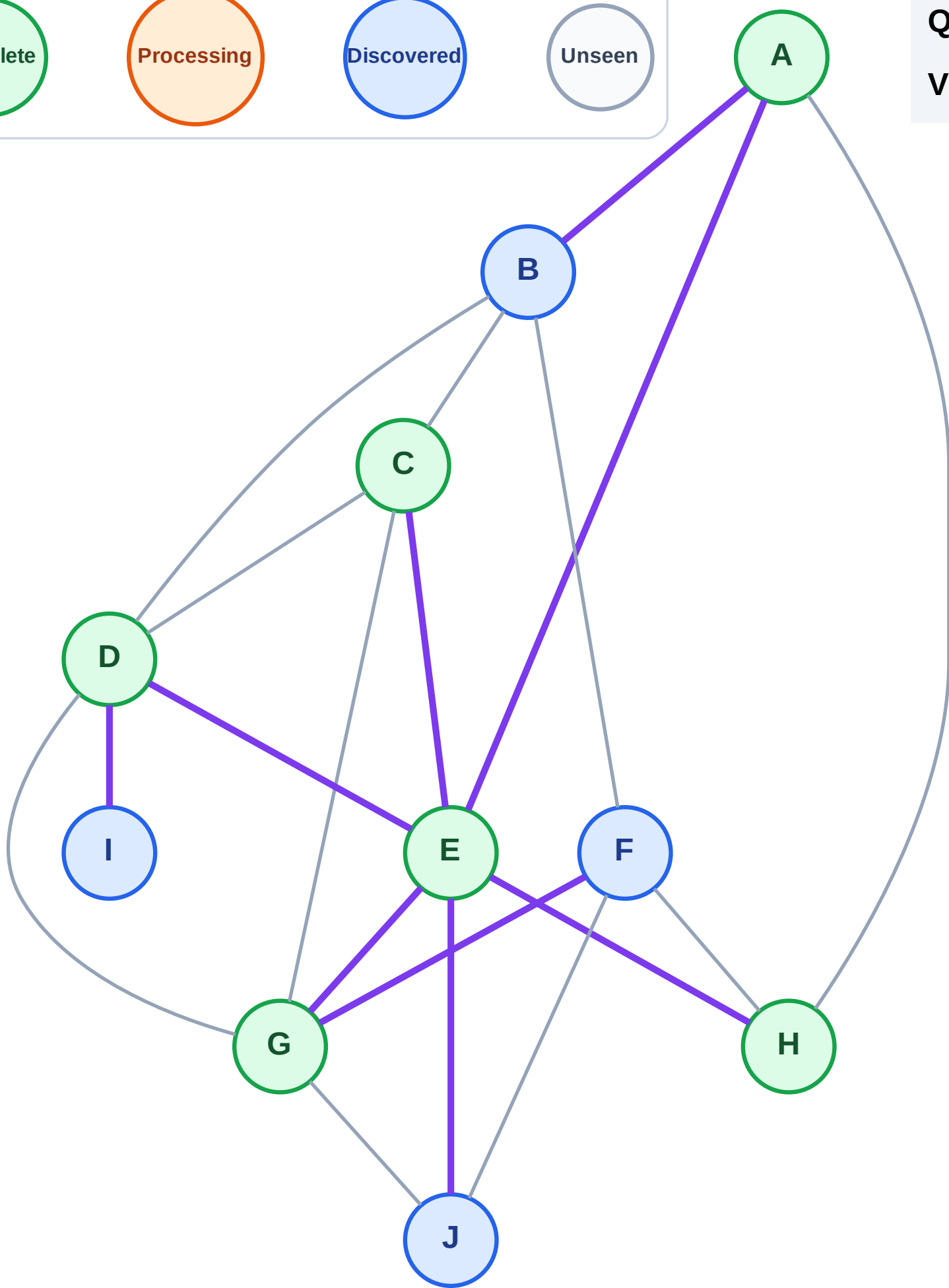
Step 13: No new neighbors from H

Legend



Queue: J → B → I → F

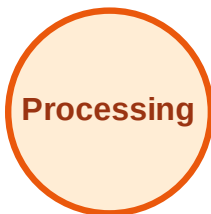
Visited: A, C, D, E, G, H



BFS Traversal

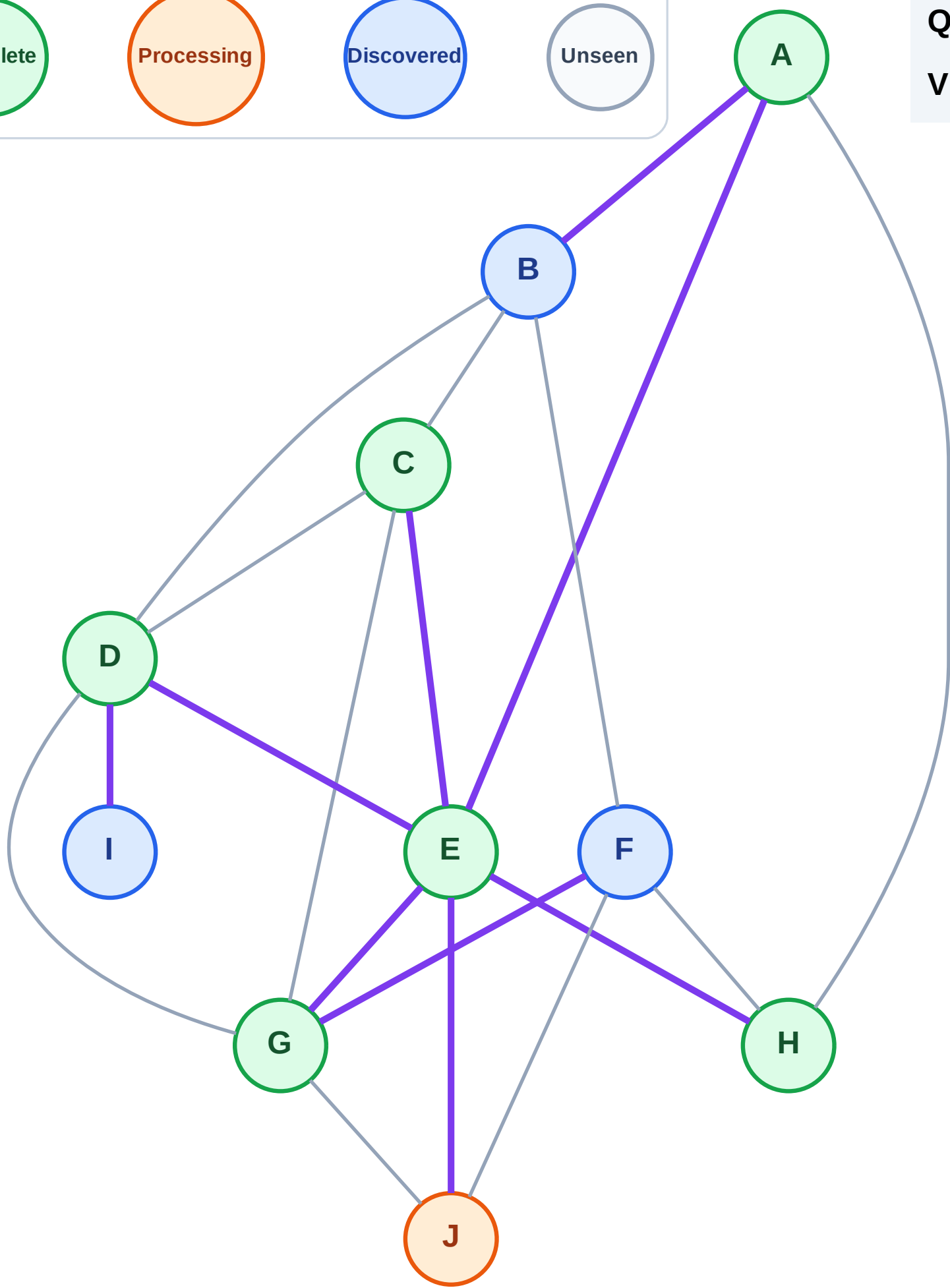
Step 14: Process node J

Legend



Queue: B → I → F

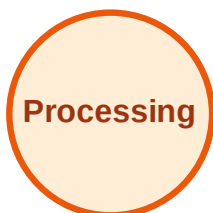
Visited: A, C, D, E, G, H



BFS Traversal

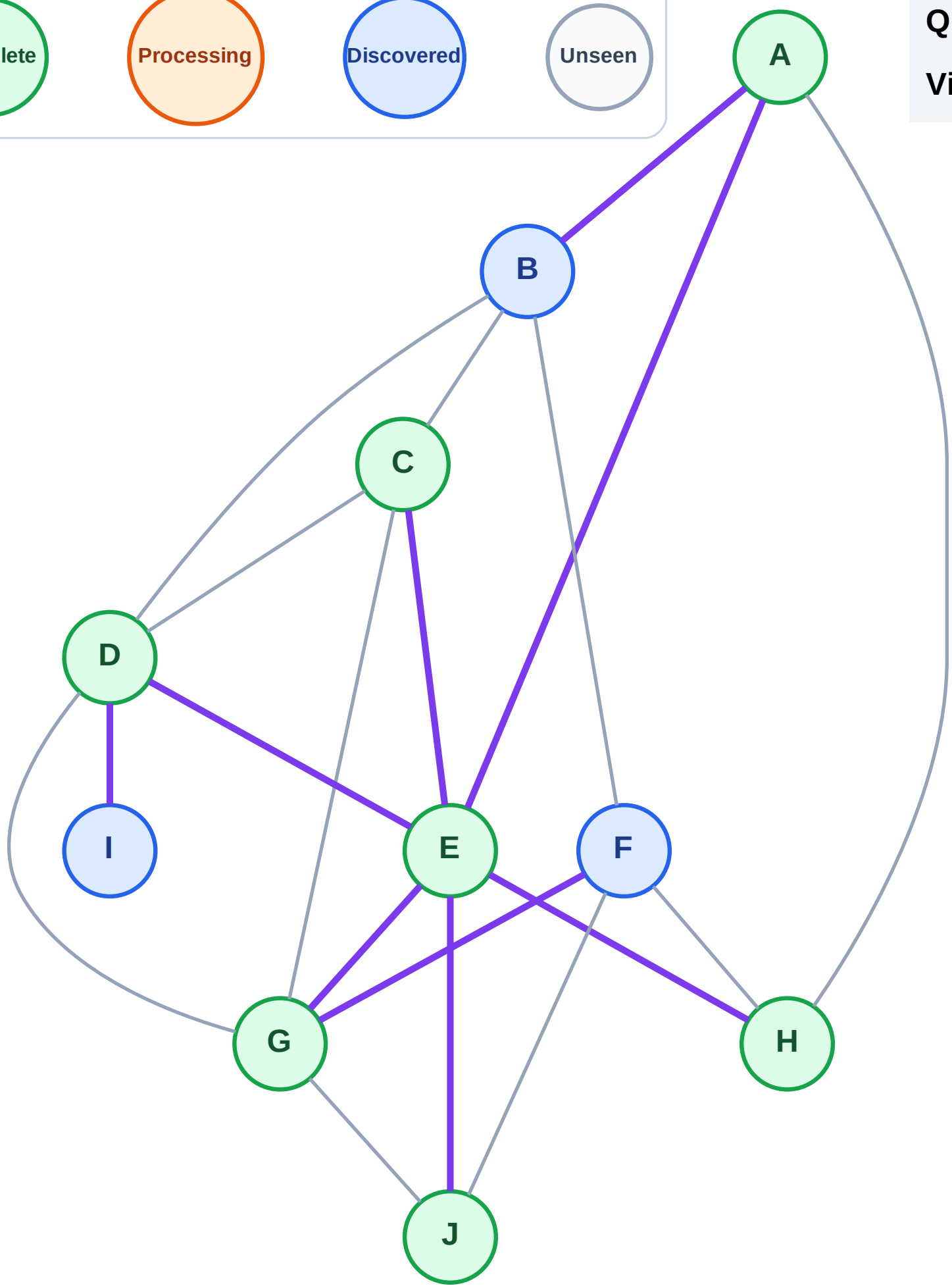
Step 15: No new neighbors from J

Legend



Queue: B → I → F

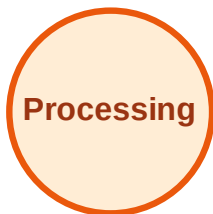
Visited: A, C, D, E, G, H, J



BFS Traversal

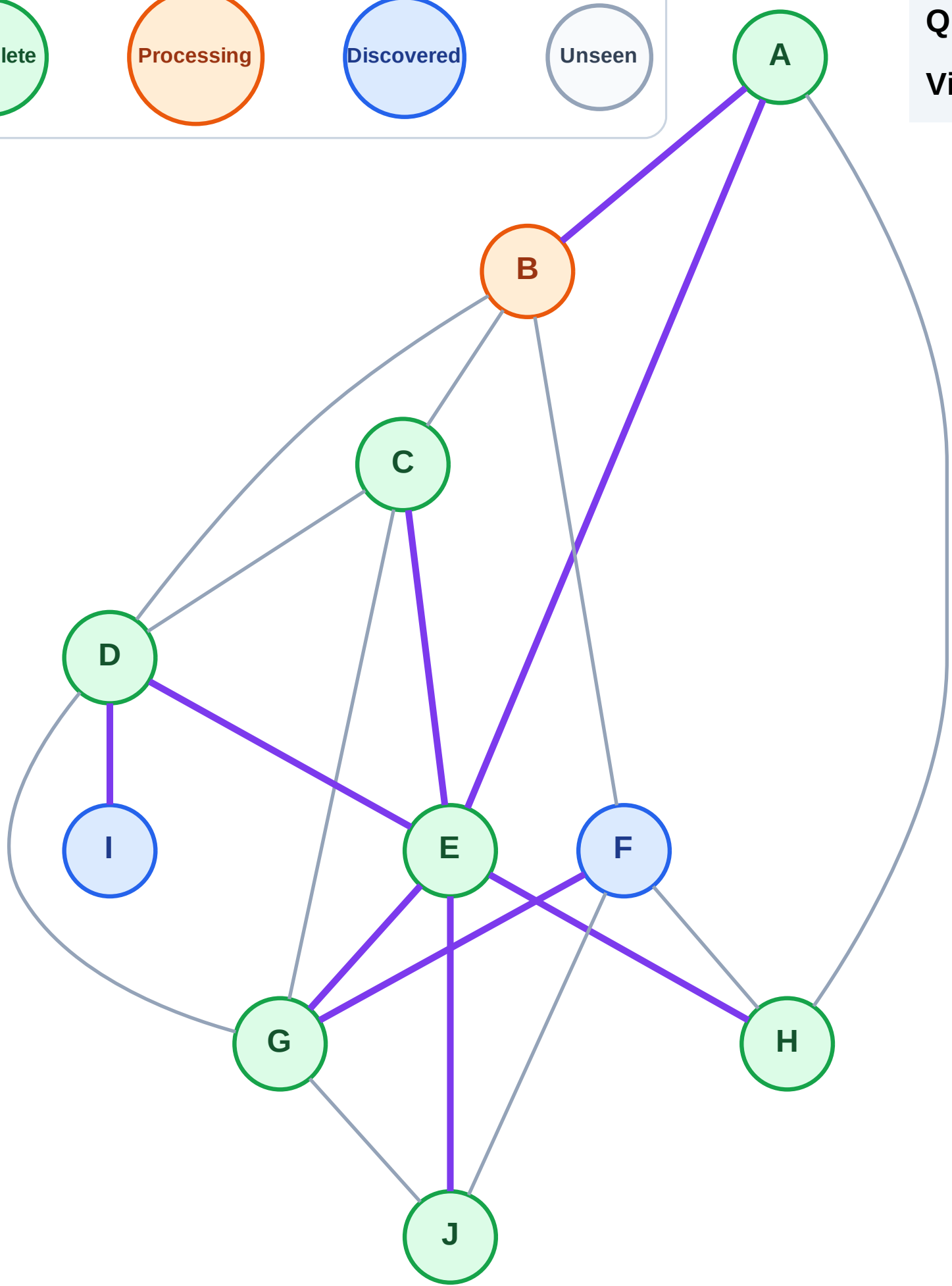
Step 16: Process node B

Legend



Queue: I → F

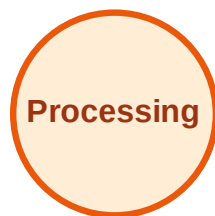
Visited: A, C, D, E, G, H, J



BFS Traversal

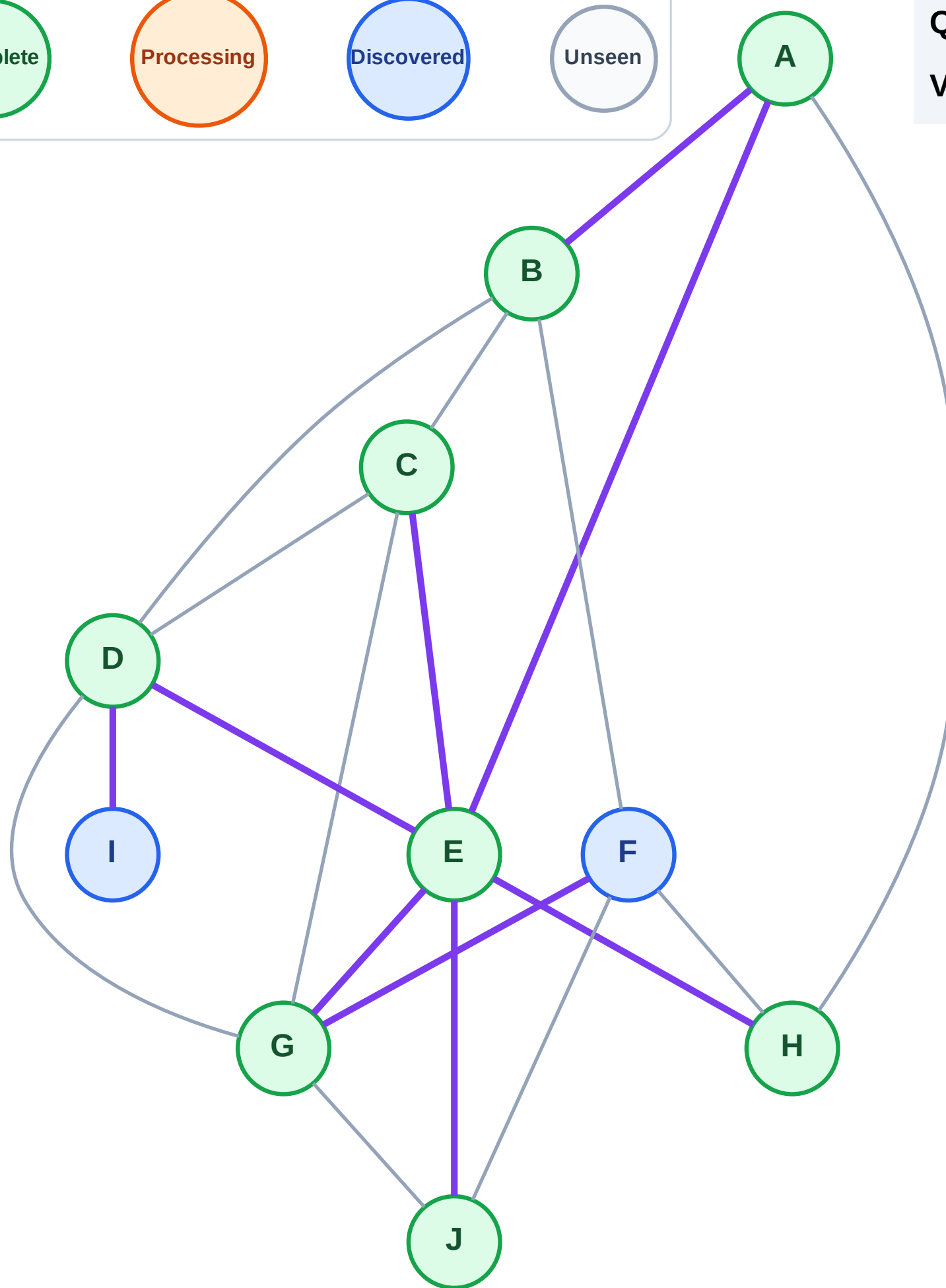
Step 17: No new neighbors from B

Legend



Queue: I → F

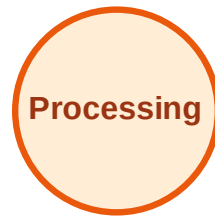
Visited: A, B, C, D, E, G, H, J



BFS Traversal

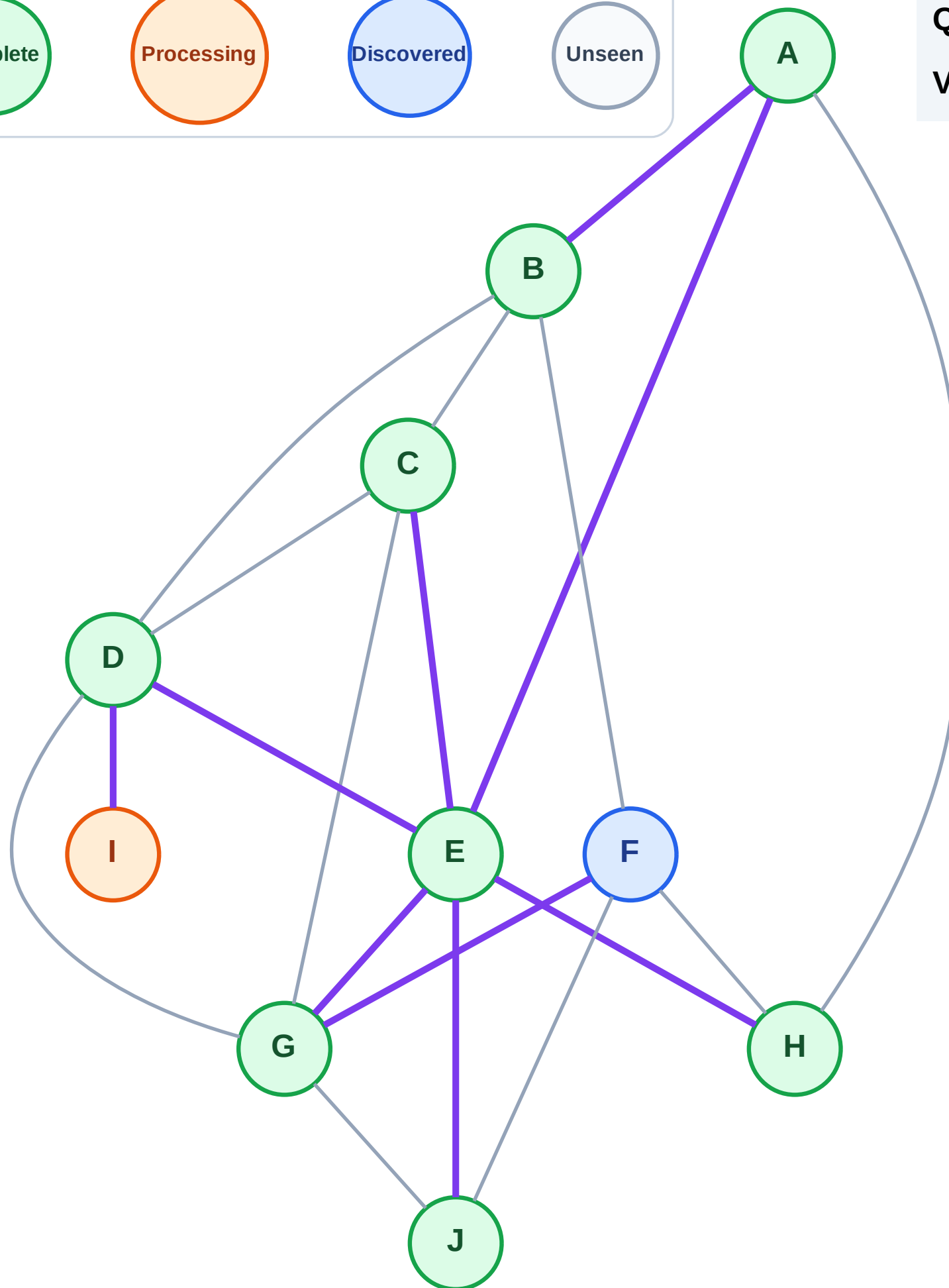
Step 18: Process node I

Legend



Queue: F

Visited: A, B, C, D, E, G, H, J



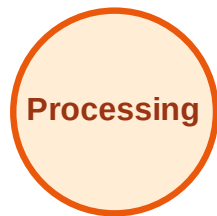
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



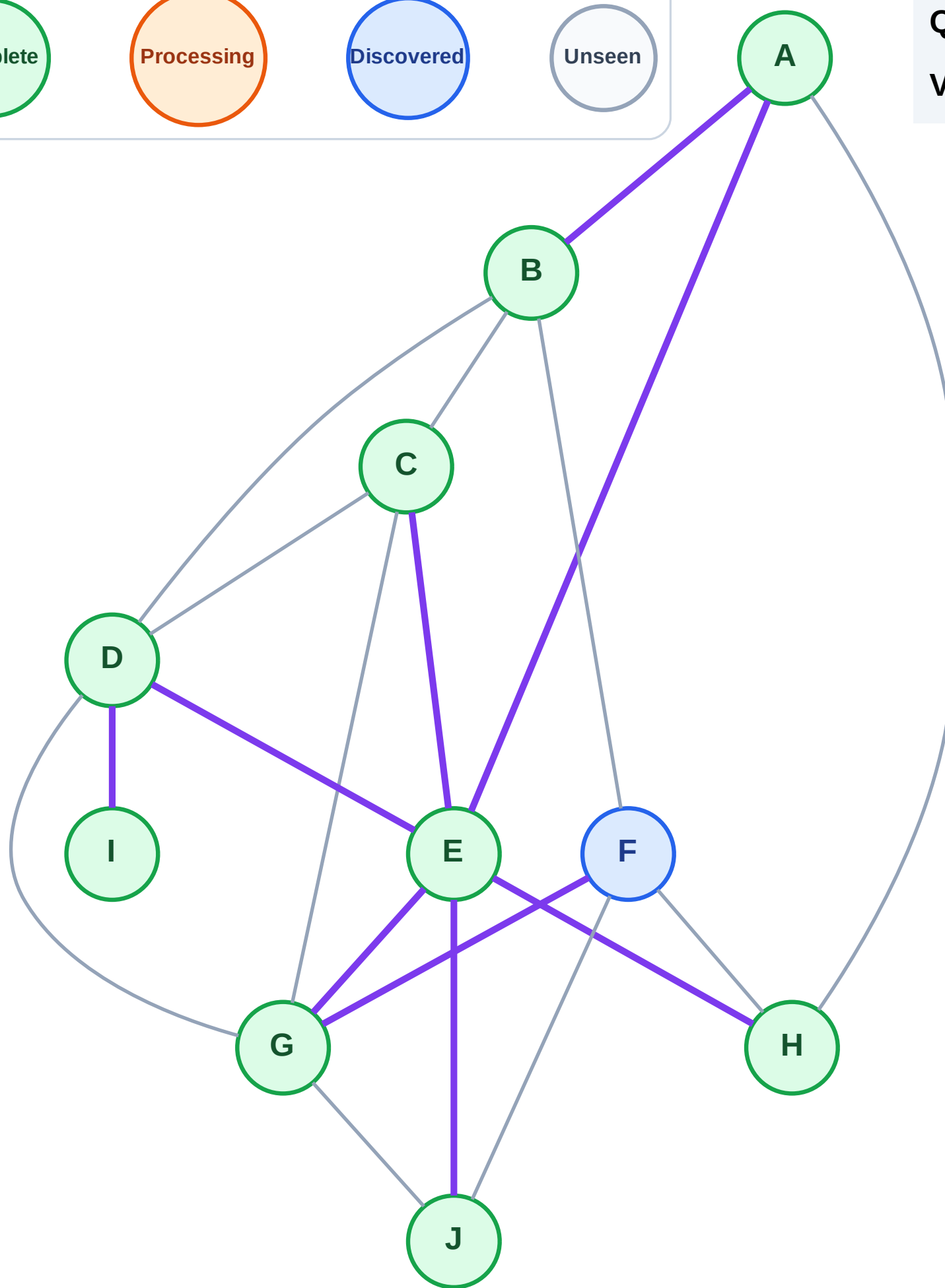
Discovered



Unseen

Queue: F

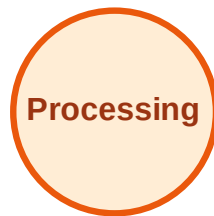
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

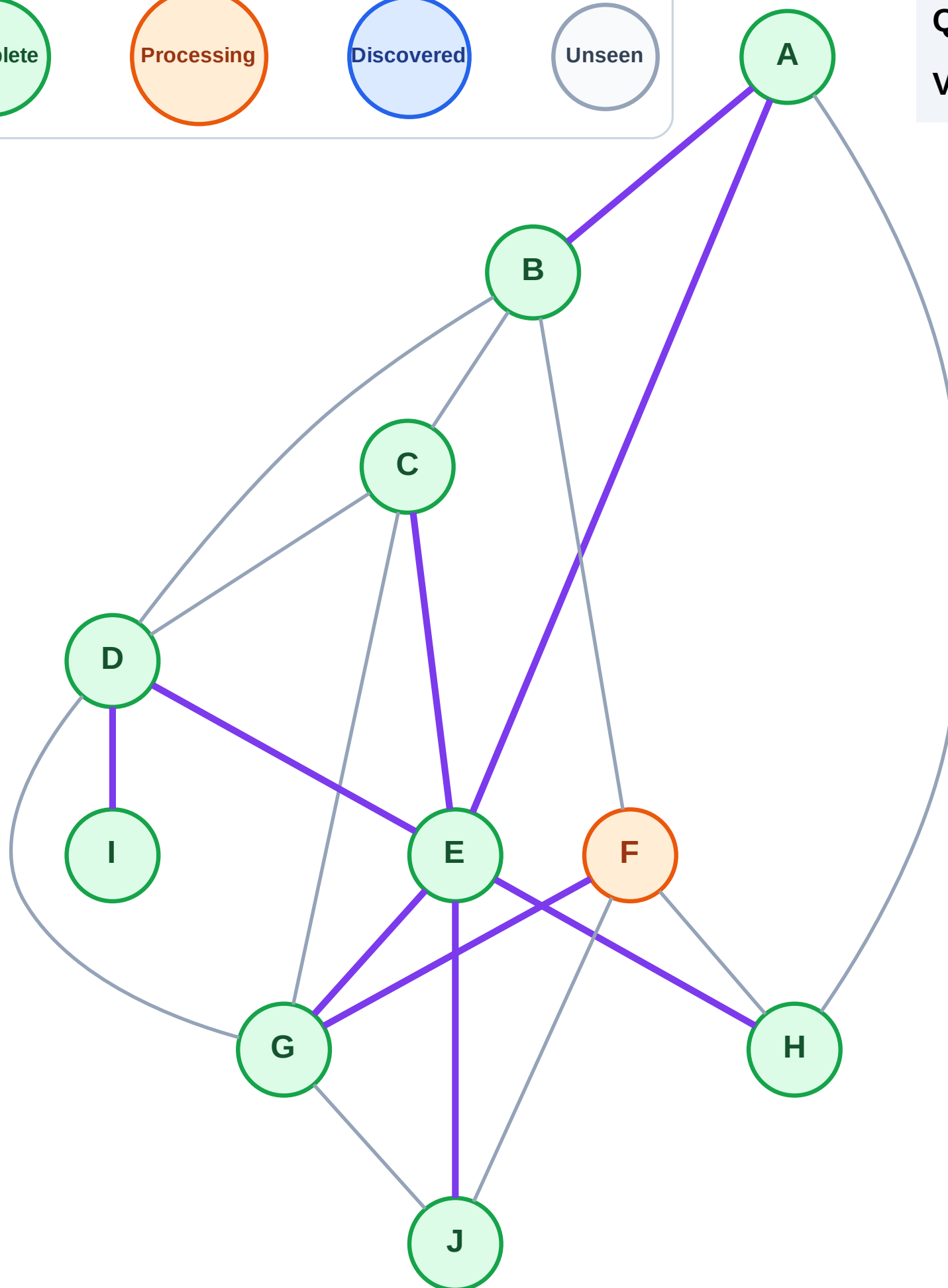
Step 20: Process node F

Legend



Queue: (empty)

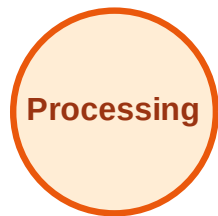
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

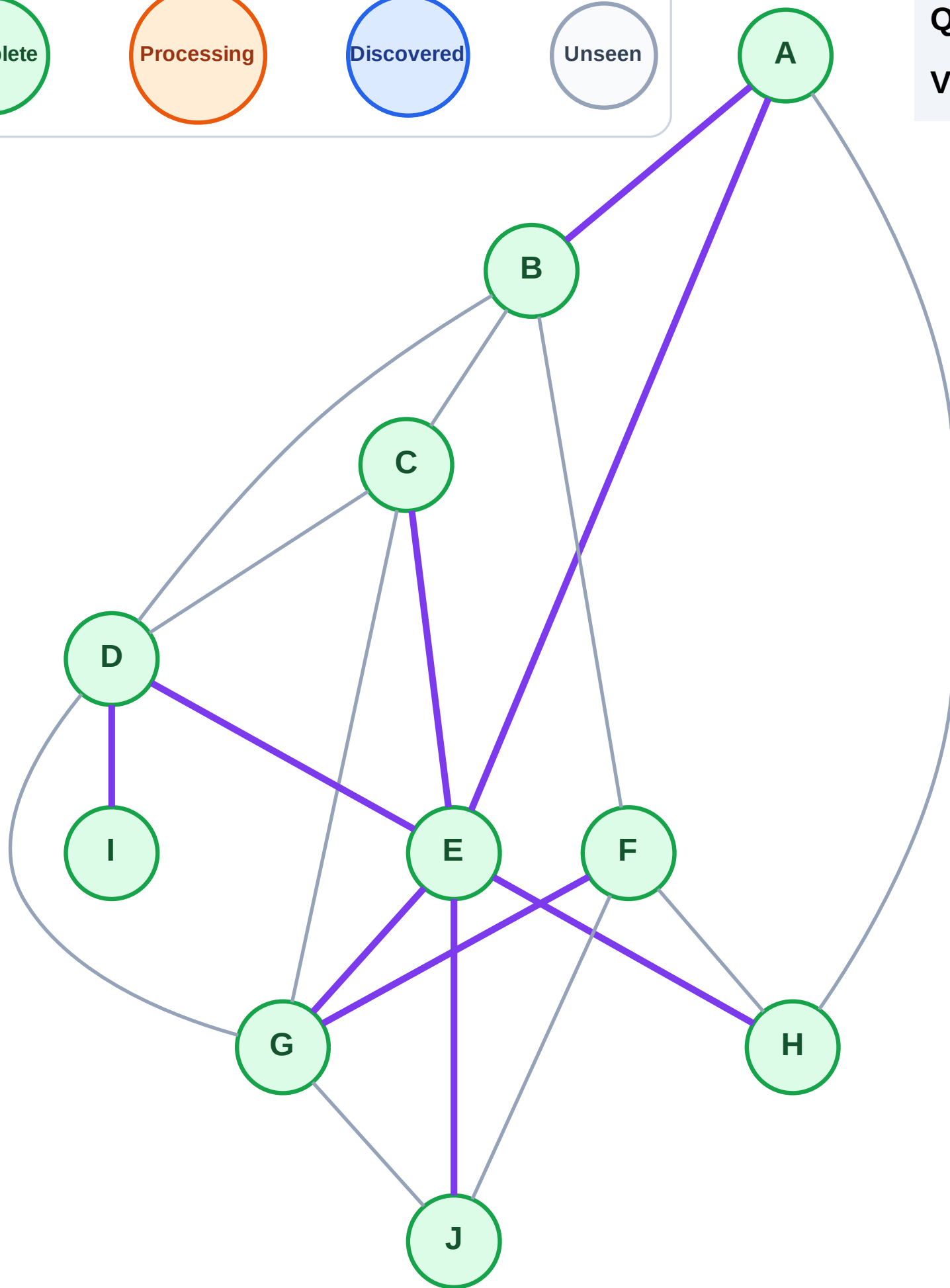
Step 21: No new neighbors from F

Legend



Queue: (empty)

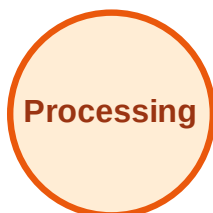
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

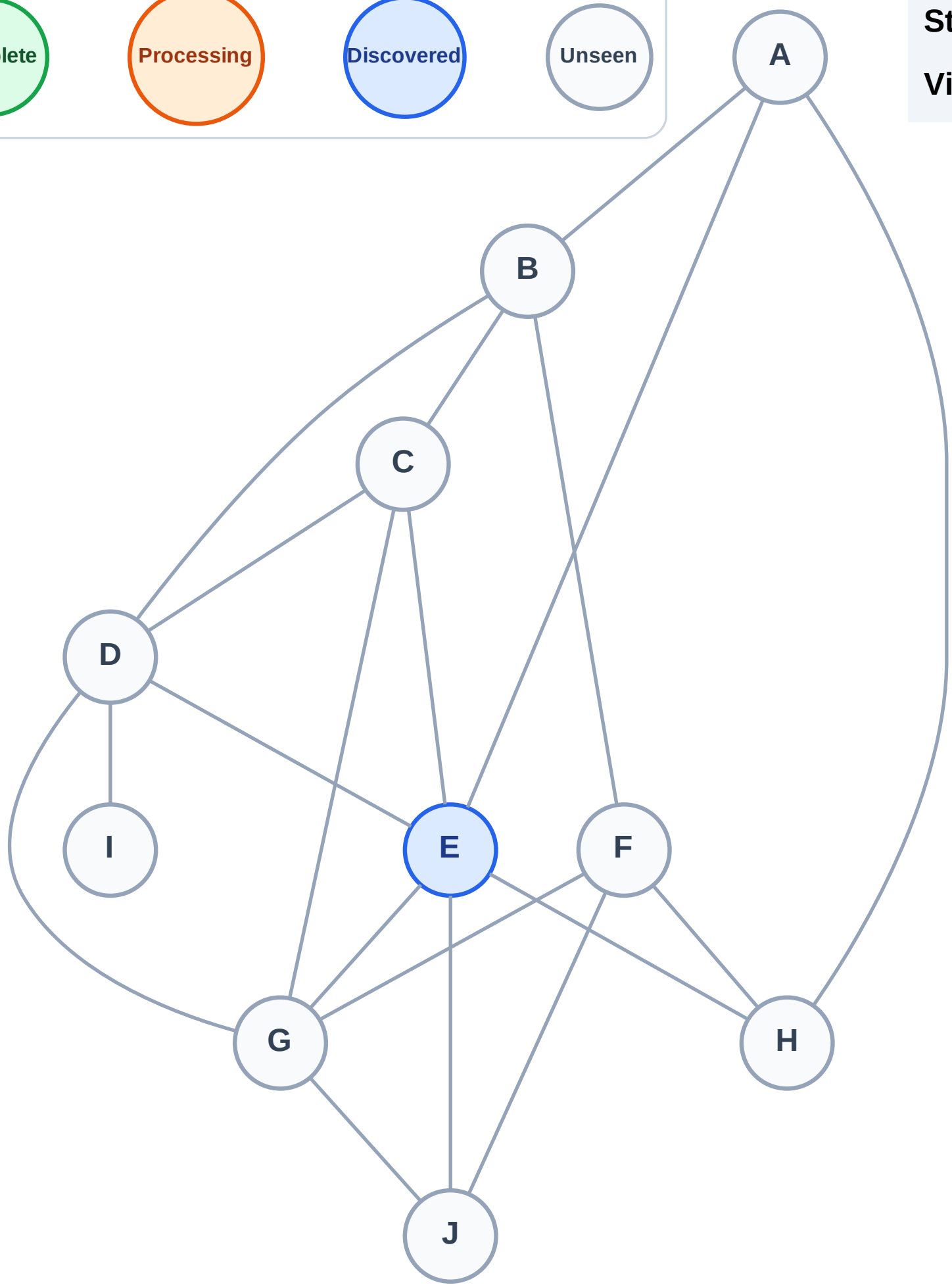
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

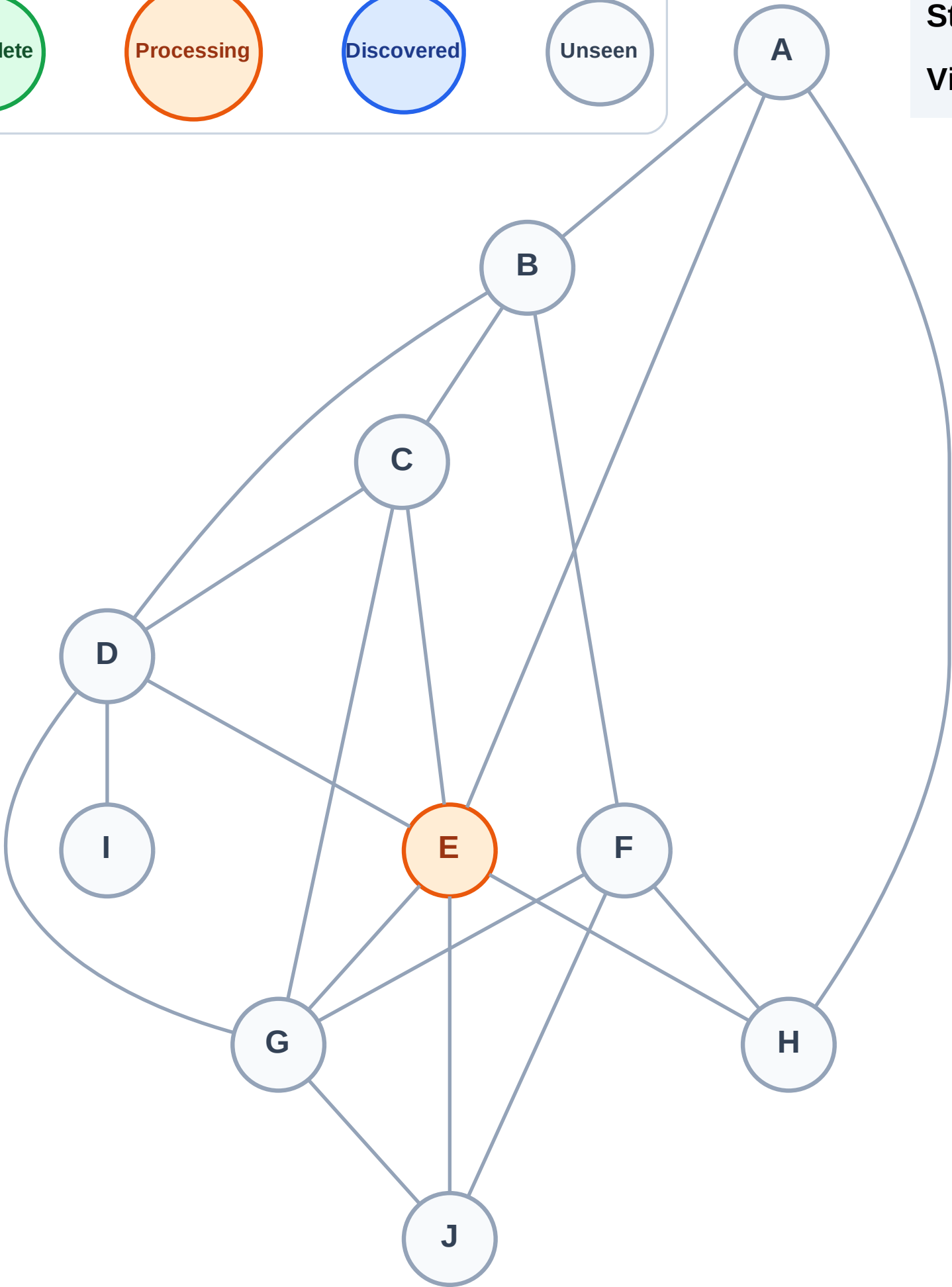
Complete

Processing

Discovered

Unseen

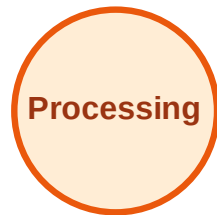
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

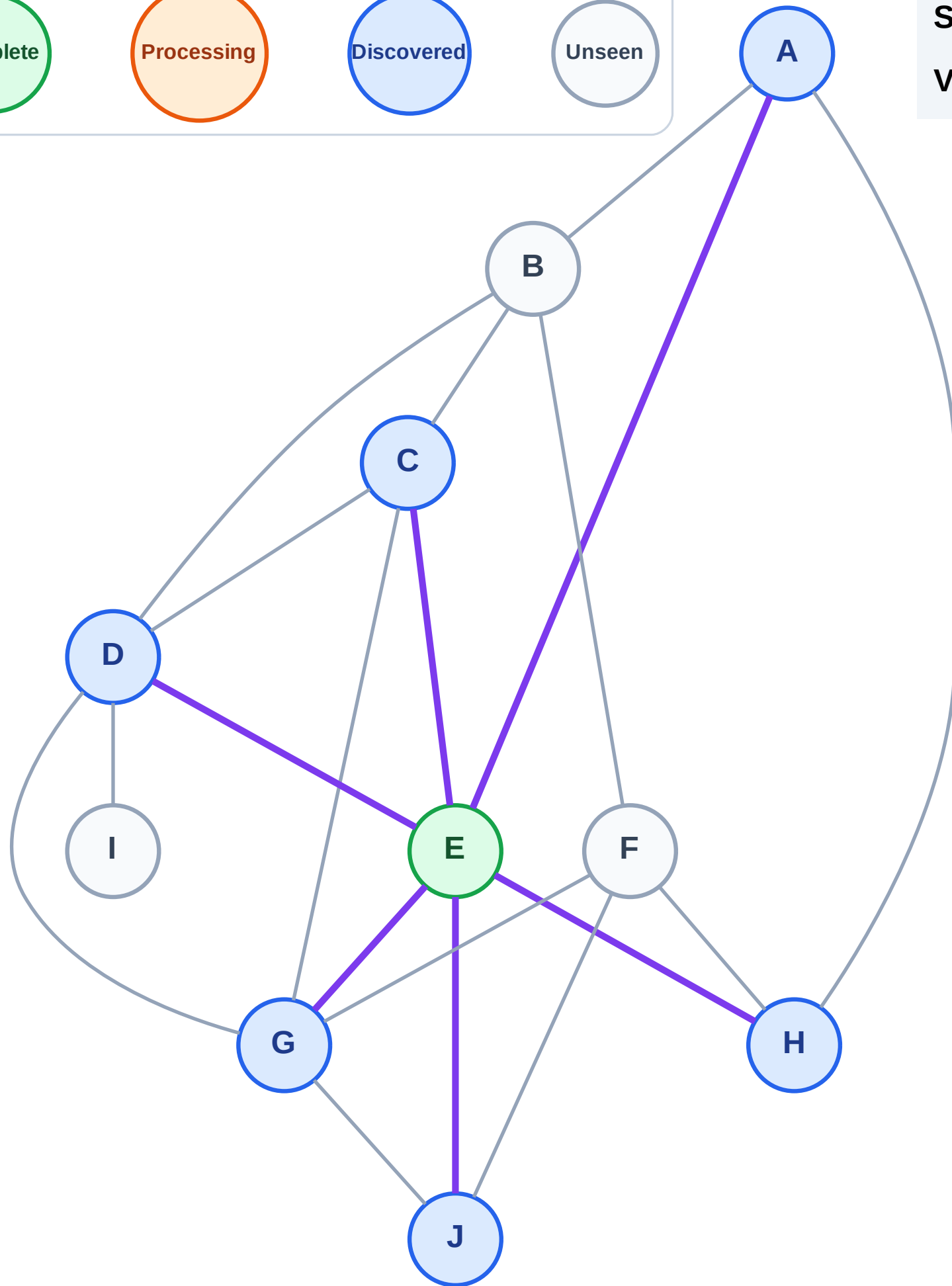
Step 3: Discover J, H, G, D, C, A from E

Legend



Stack (top at right): J | H | G | D | C | A

Visited: E



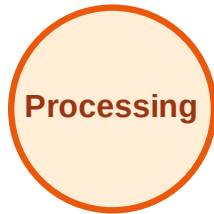
DFS Traversal

Step 4: Process node A

Legend



Complete



Processing



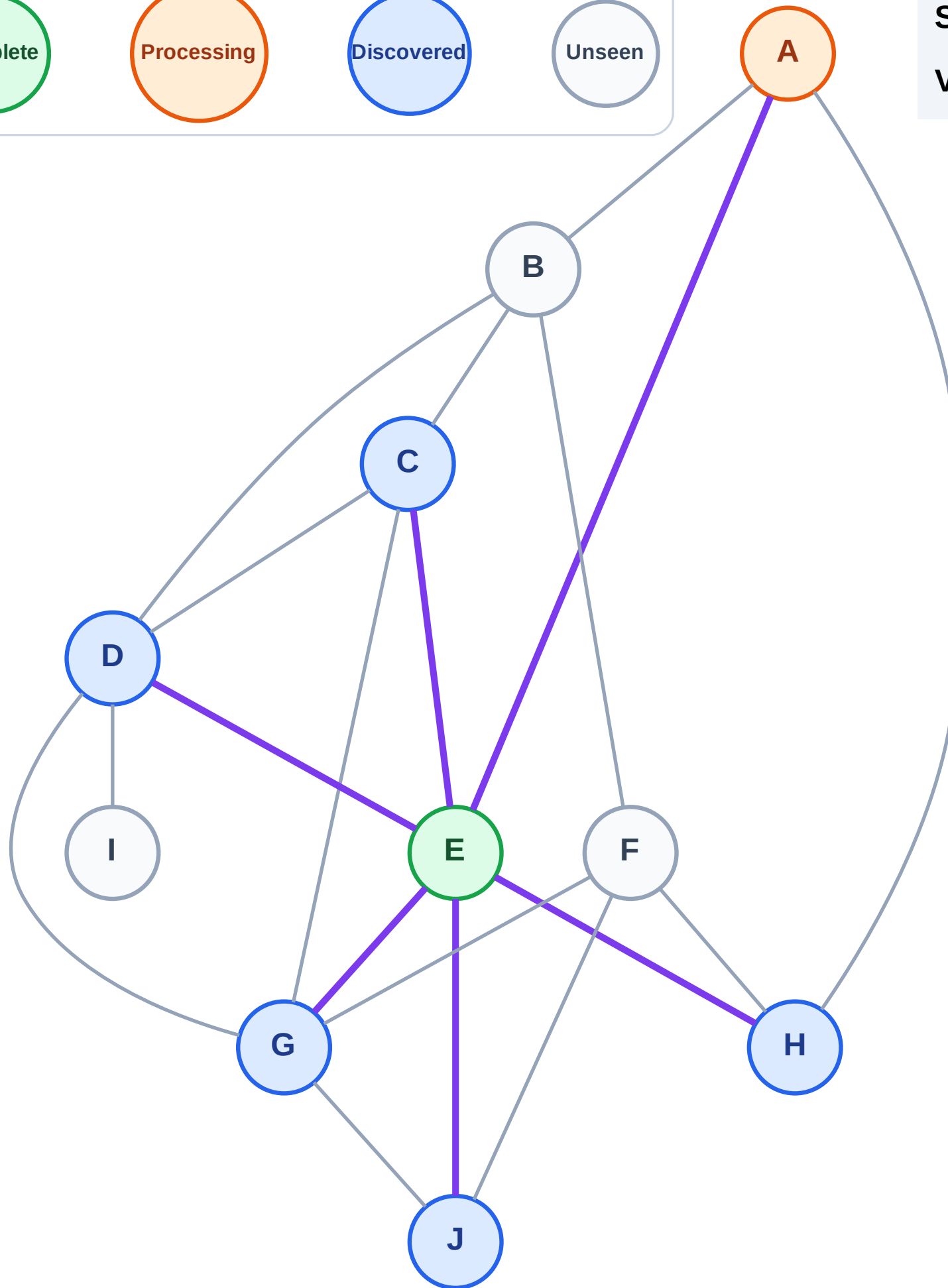
Discovered



Unseen

Stack (top at right): J | H | G | D | C

Visited: E



DFS Traversal

Step 5: Discover B from A

Legend

Complete

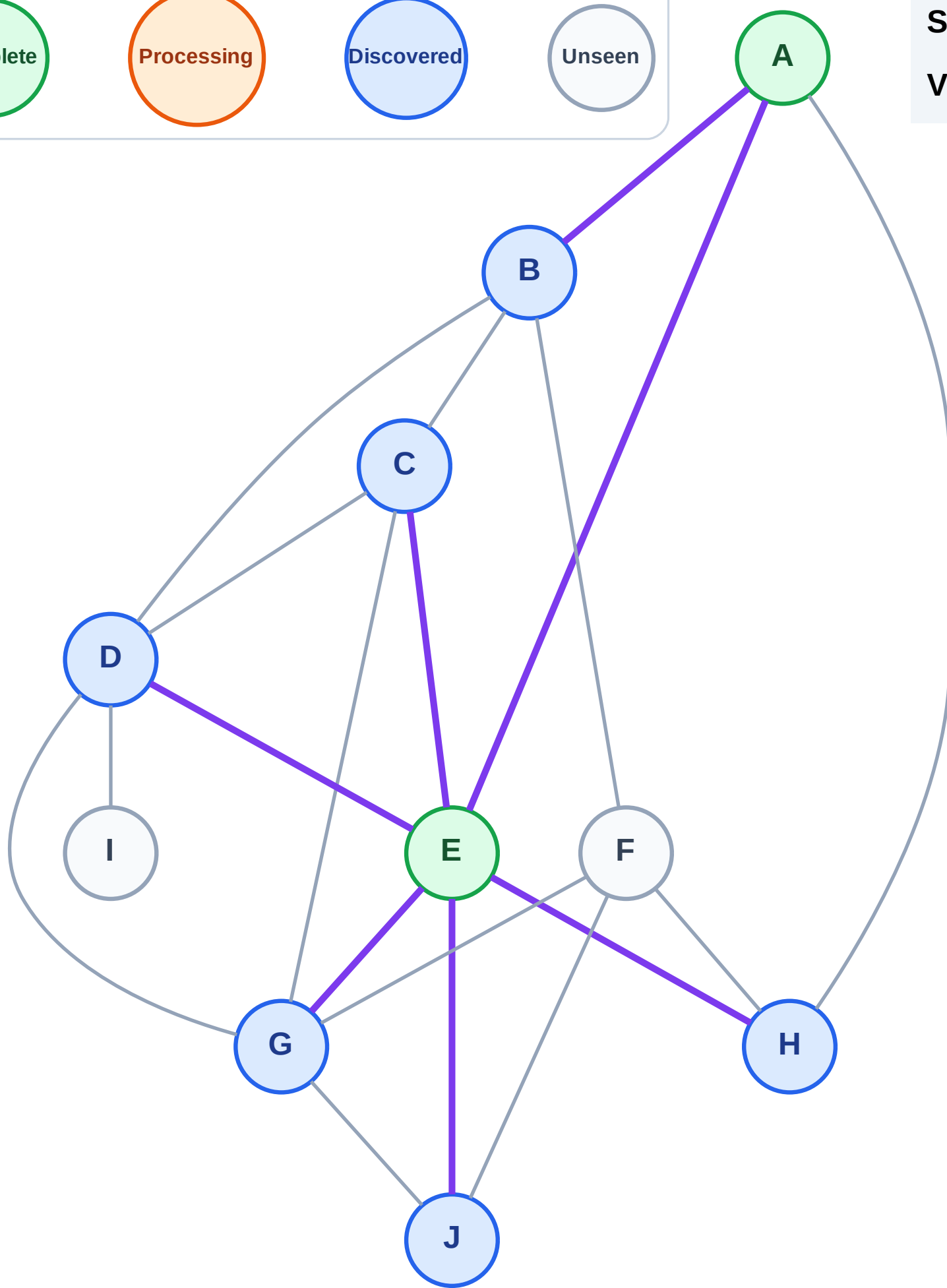
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D | C | B

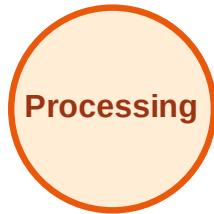
Visited: A, E



DFS Traversal

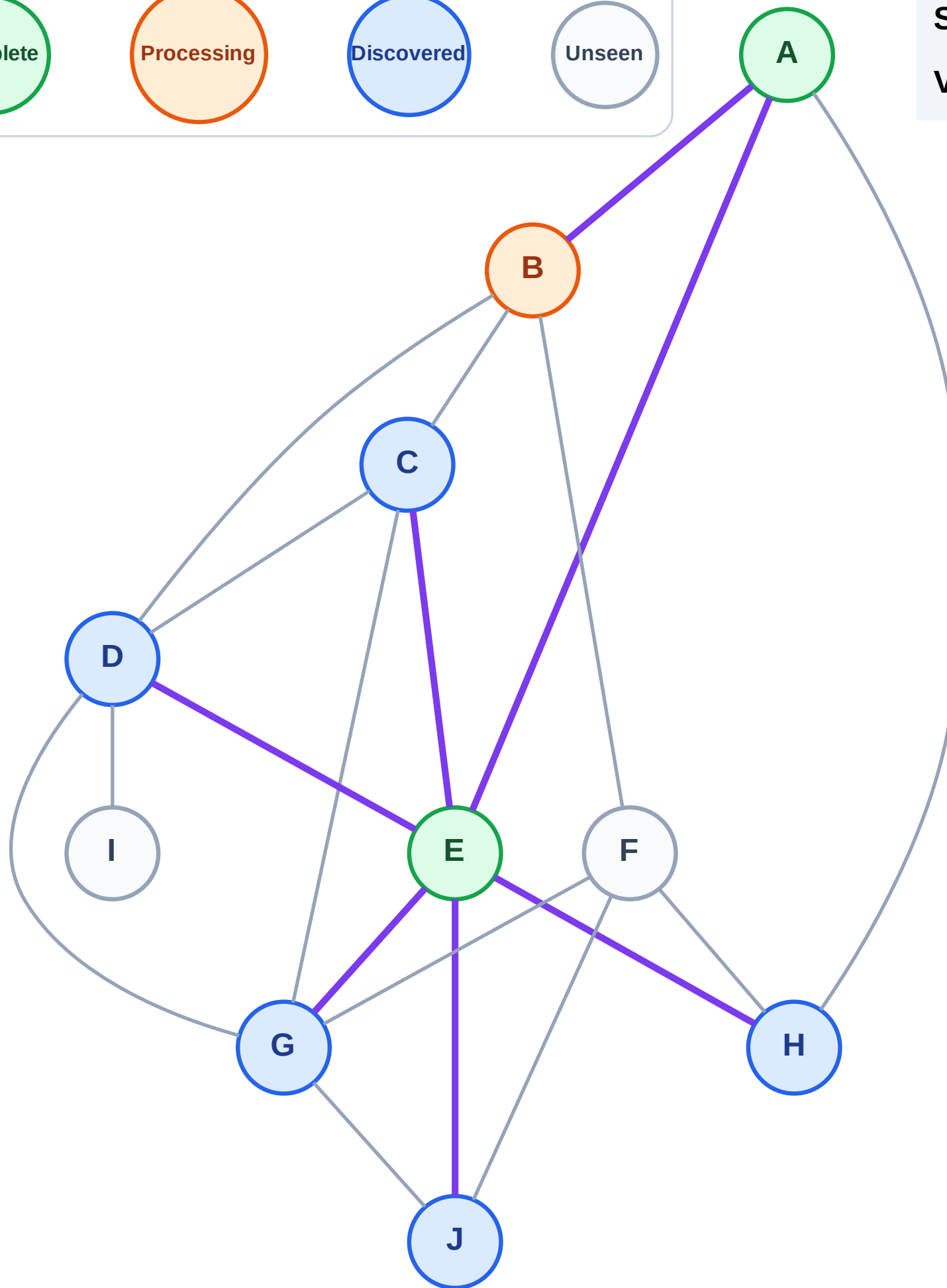
Step 6: Process node B

Legend



Stack (top at right): J | H | G | D | C

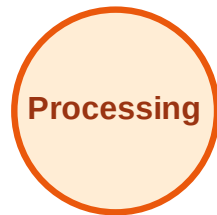
Visited: A, E



DFS Traversal

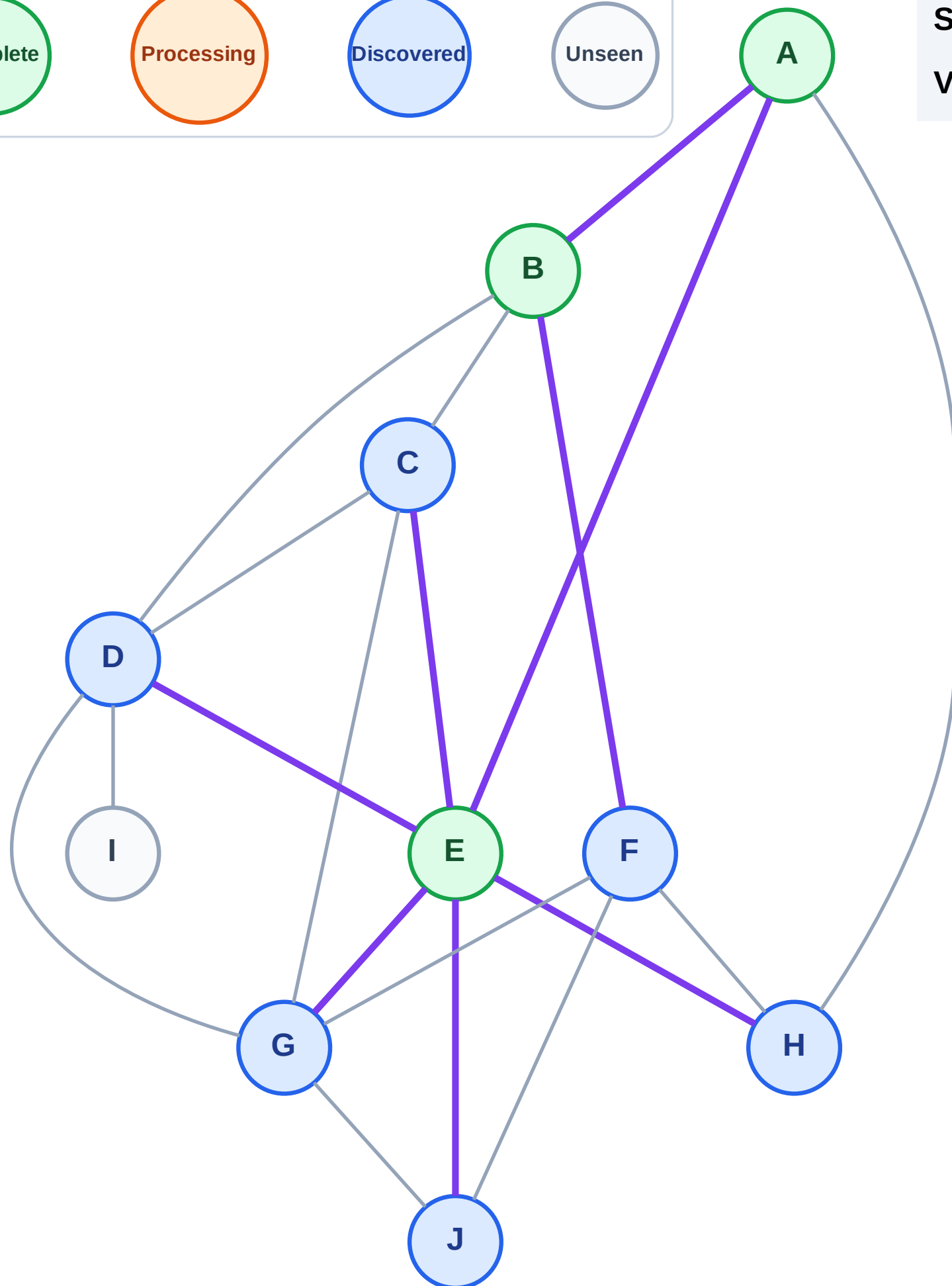
Step 7: Discover F from B

Legend



Stack (top at right): J | H | G | D | C | F

Visited: A, B, E



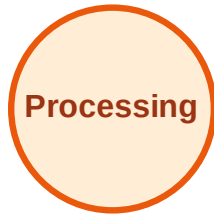
DFS Traversal

Step 8: Process node F

Legend



Complete



Processing



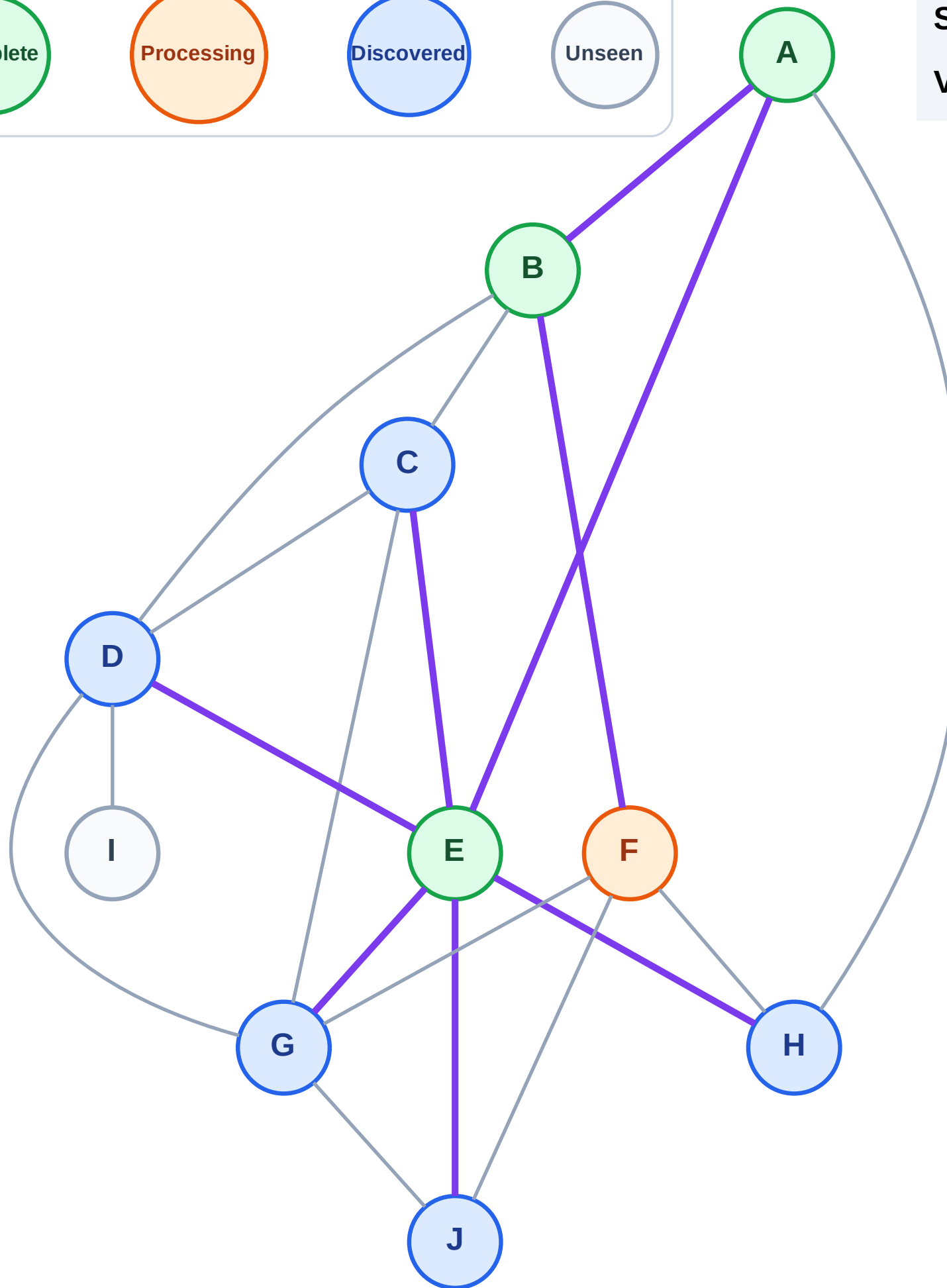
Discovered



Unseen

Stack (top at right): J | H | G | D | C

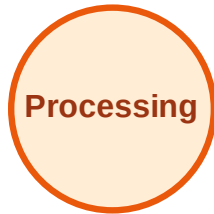
Visited: A, B, E



DFS Traversal

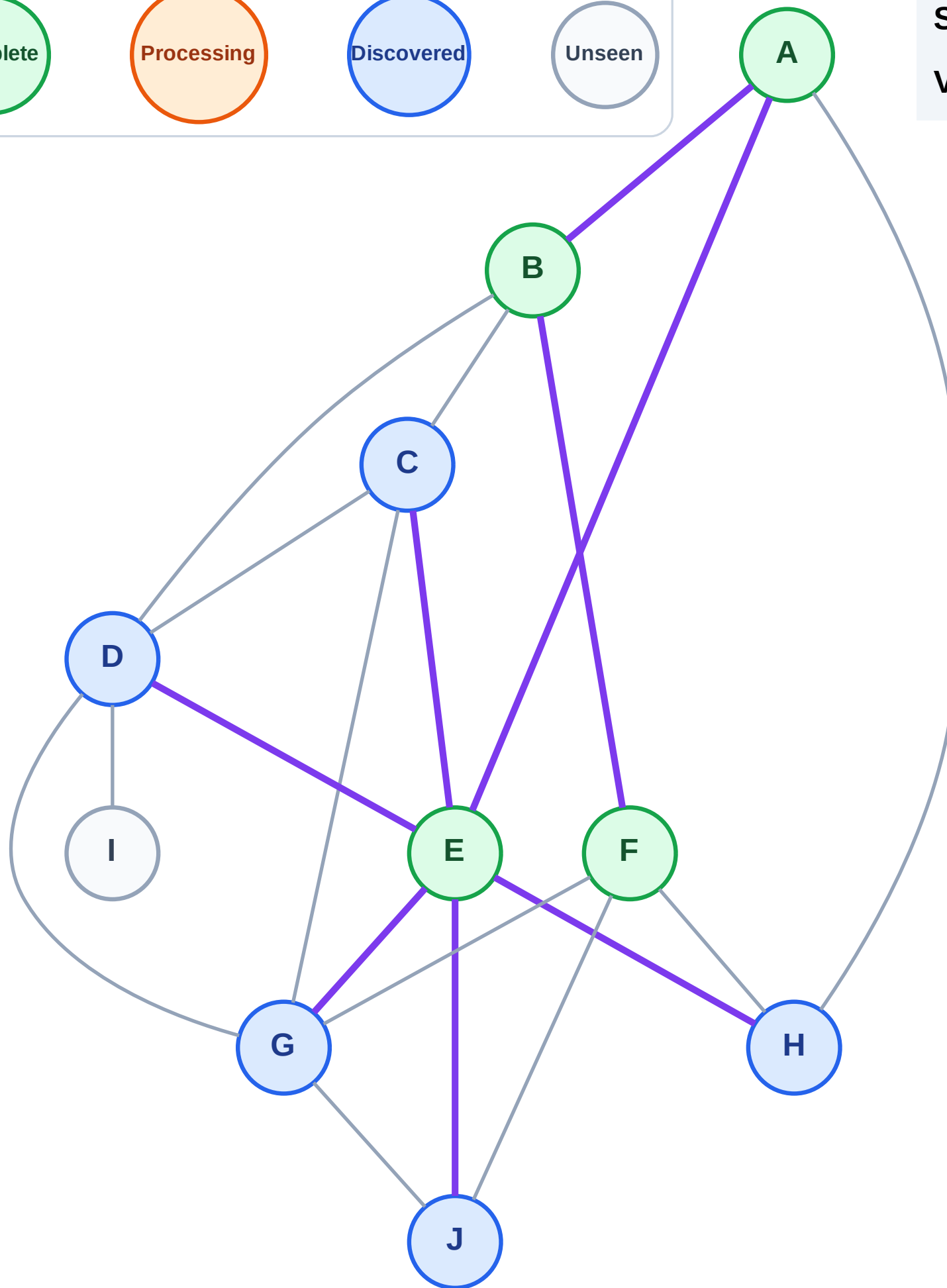
Step 9: No new neighbors from F

Legend



Stack (top at right): J | H | G | D | C

Visited: A, B, E, F



Step 10: Process node C



DFS Traversal

Step 11: No new neighbors from C

Legend

Complete

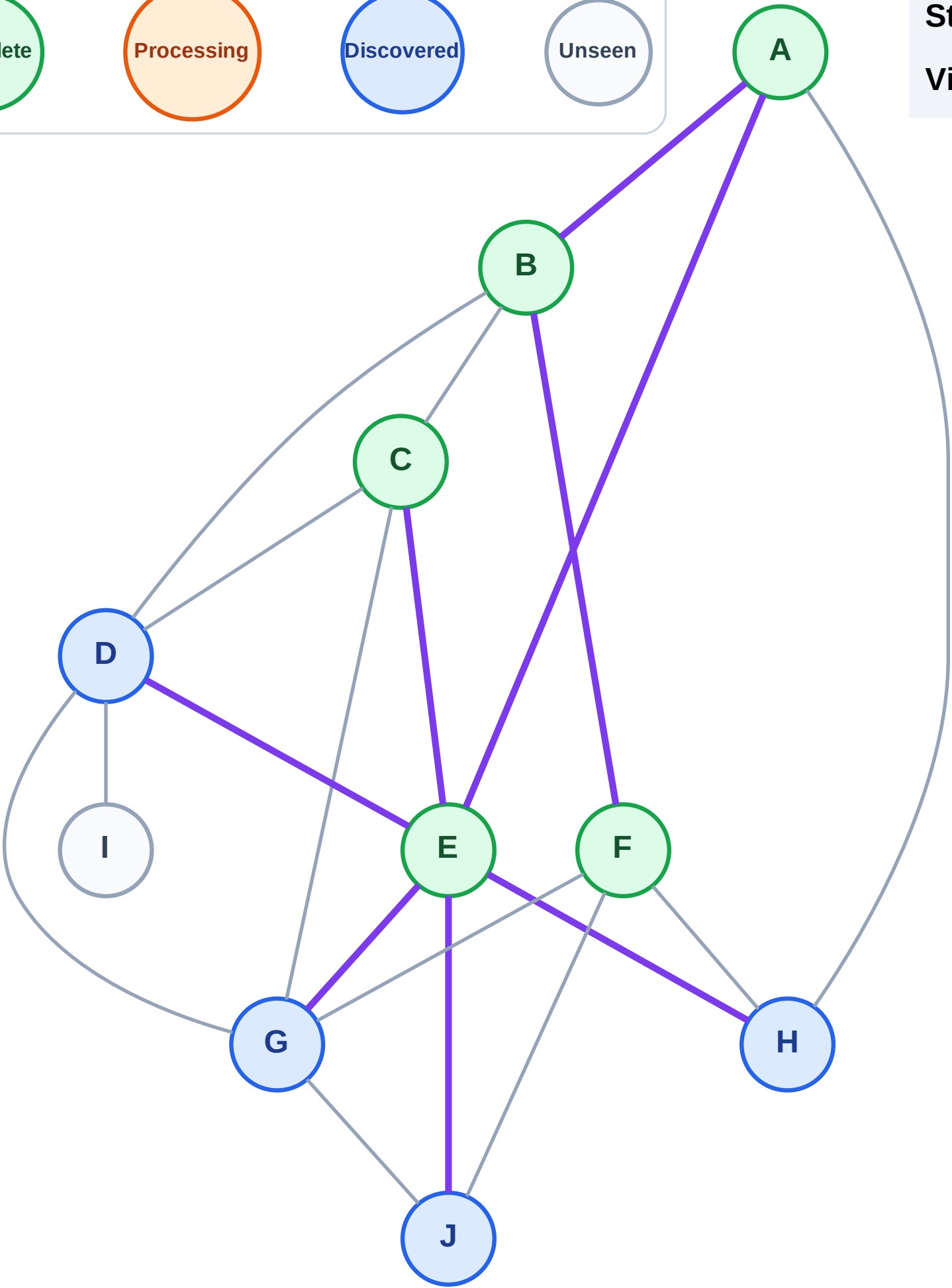
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D

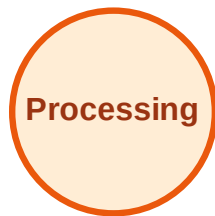
Visited: A, B, C, E, F



DFS Traversal

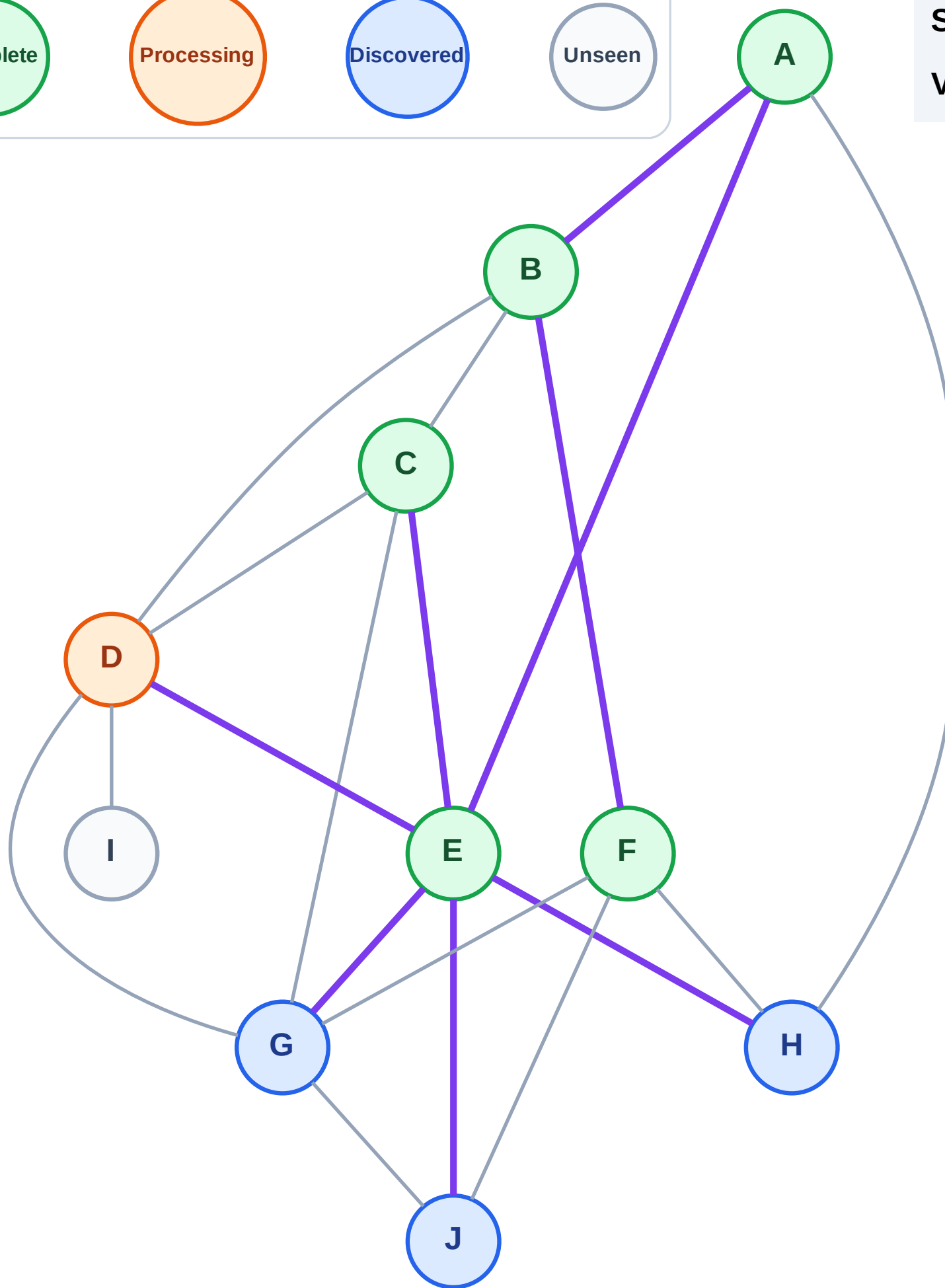
Step 12: Process node D

Legend



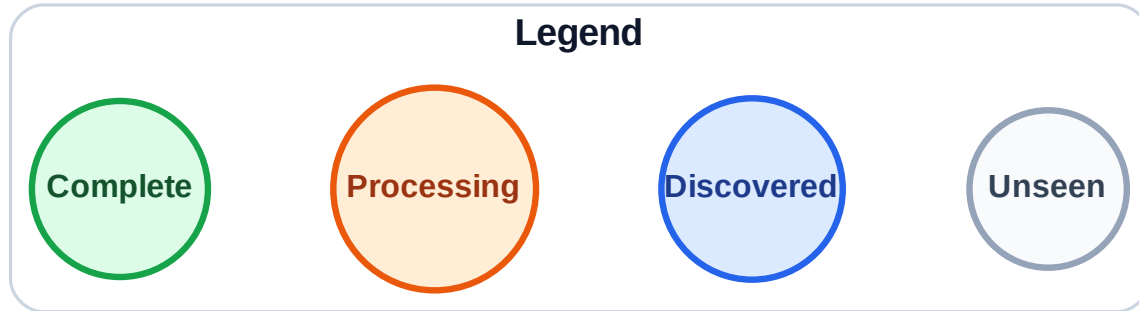
Stack (top at right): J | H | G

Visited: A, B, C, E, F



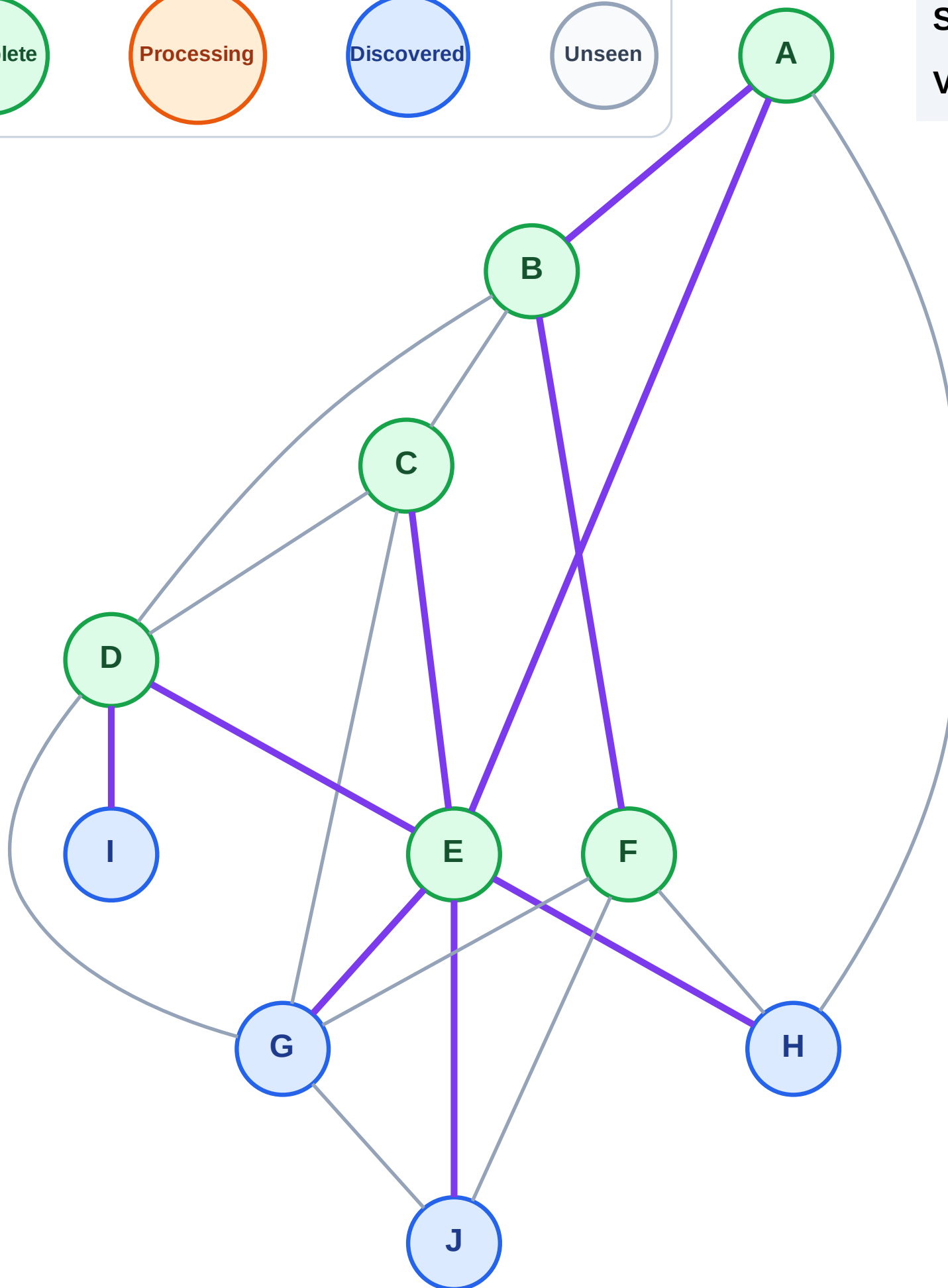
Step 13: Discover I from D

Step 13: Discover I from D



Stack (top at right): J | H | G | I

Visited: A, B, C, D, E, F



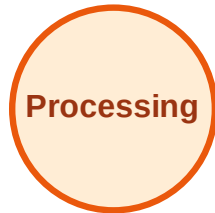
DFS Traversal

Step 14: Process node I

Legend



Complete



Processing



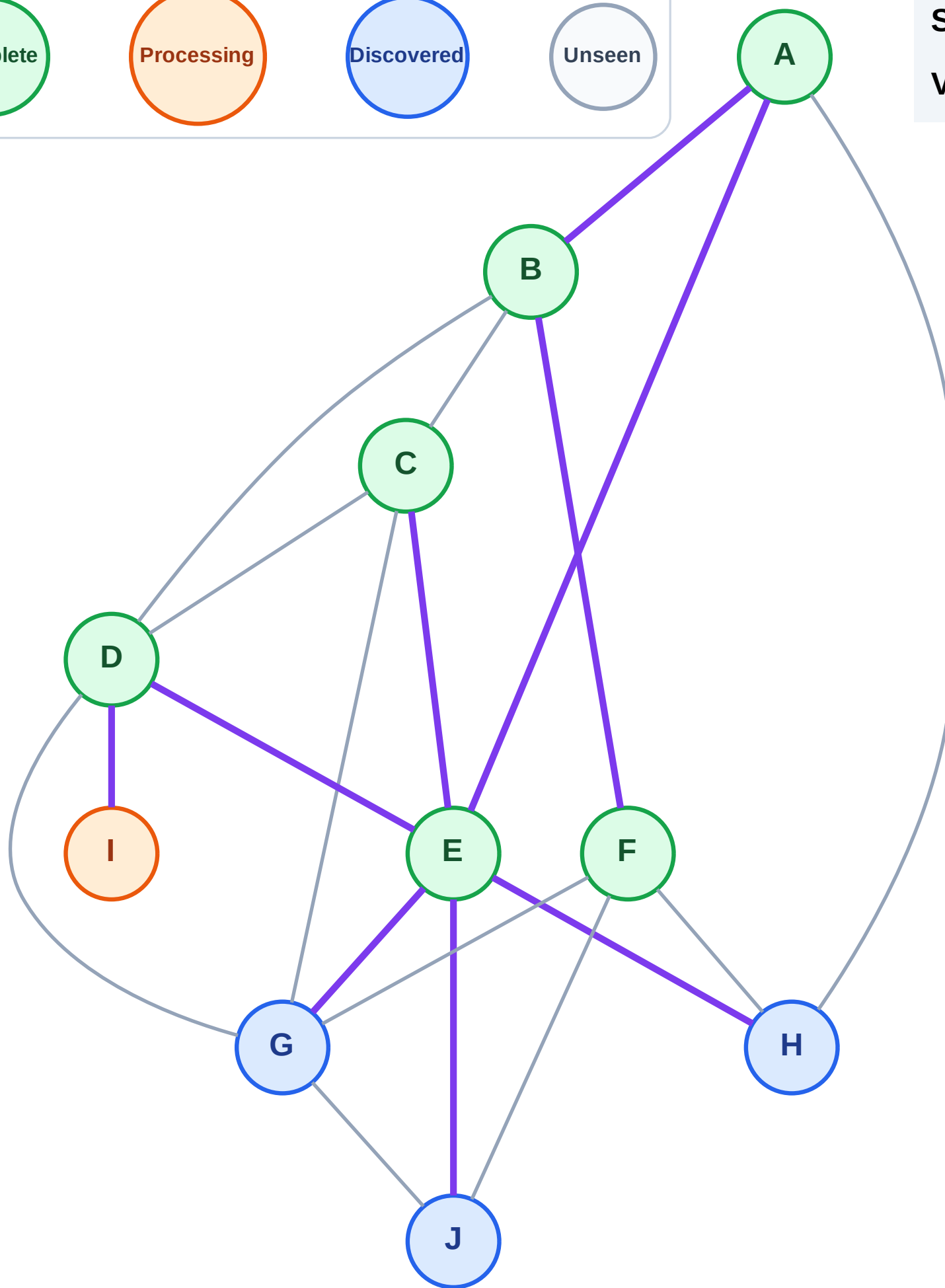
Discovered



Unseen

Stack (top at right): J | H | G

Visited: A, B, C, D, E, F



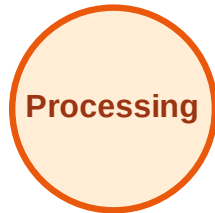
DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



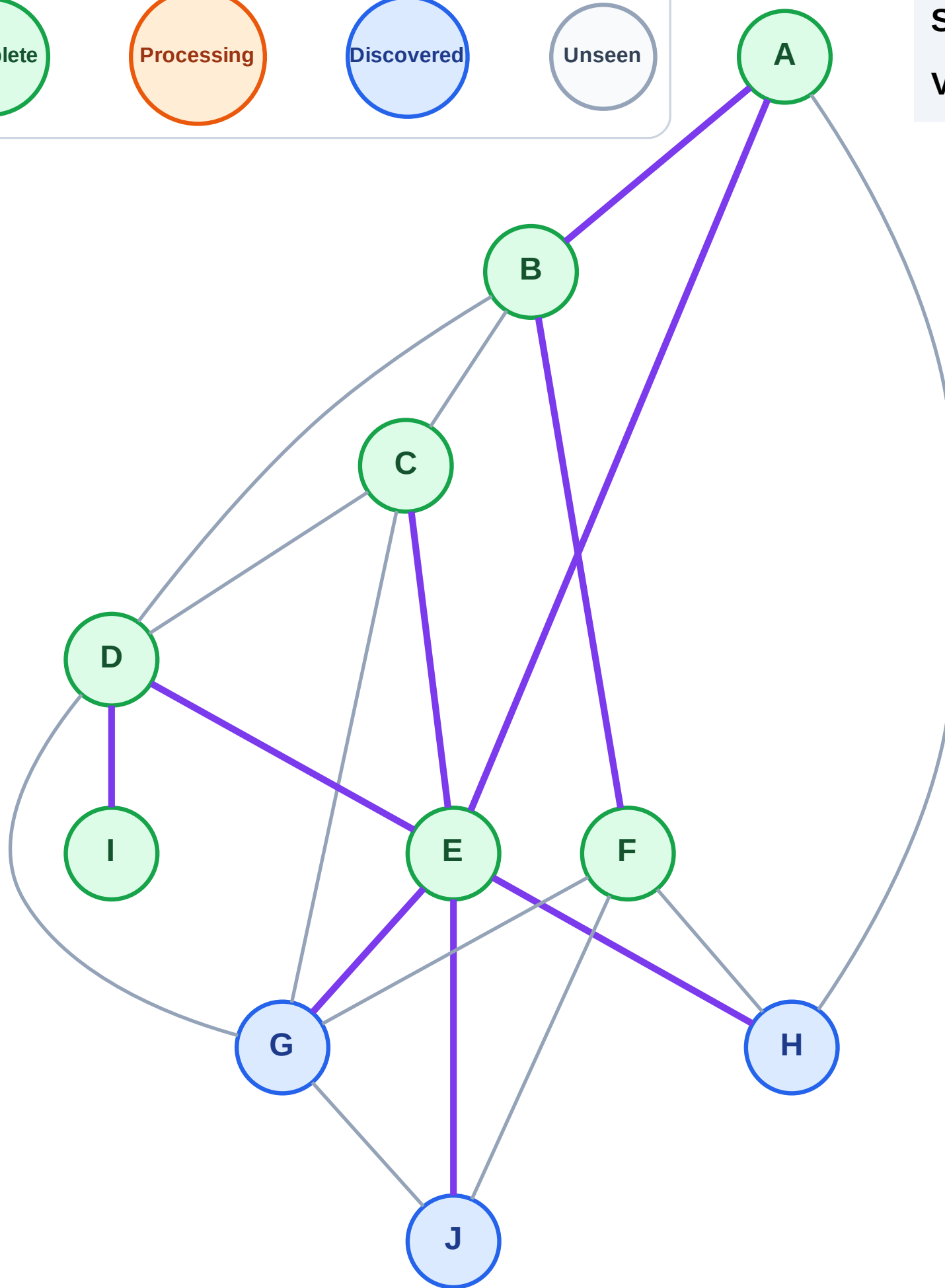
Discovered



Unseen

Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

Complete

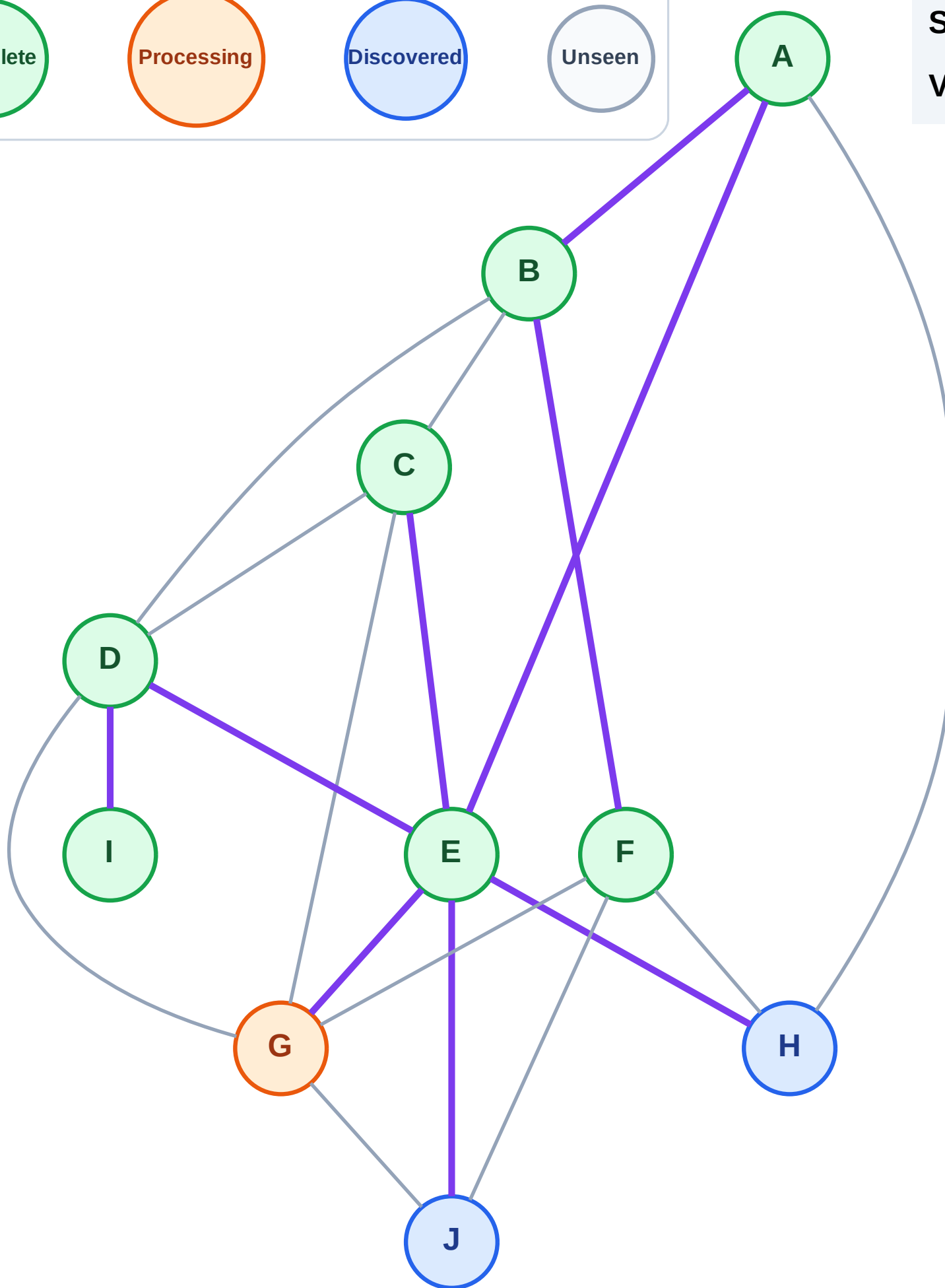
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

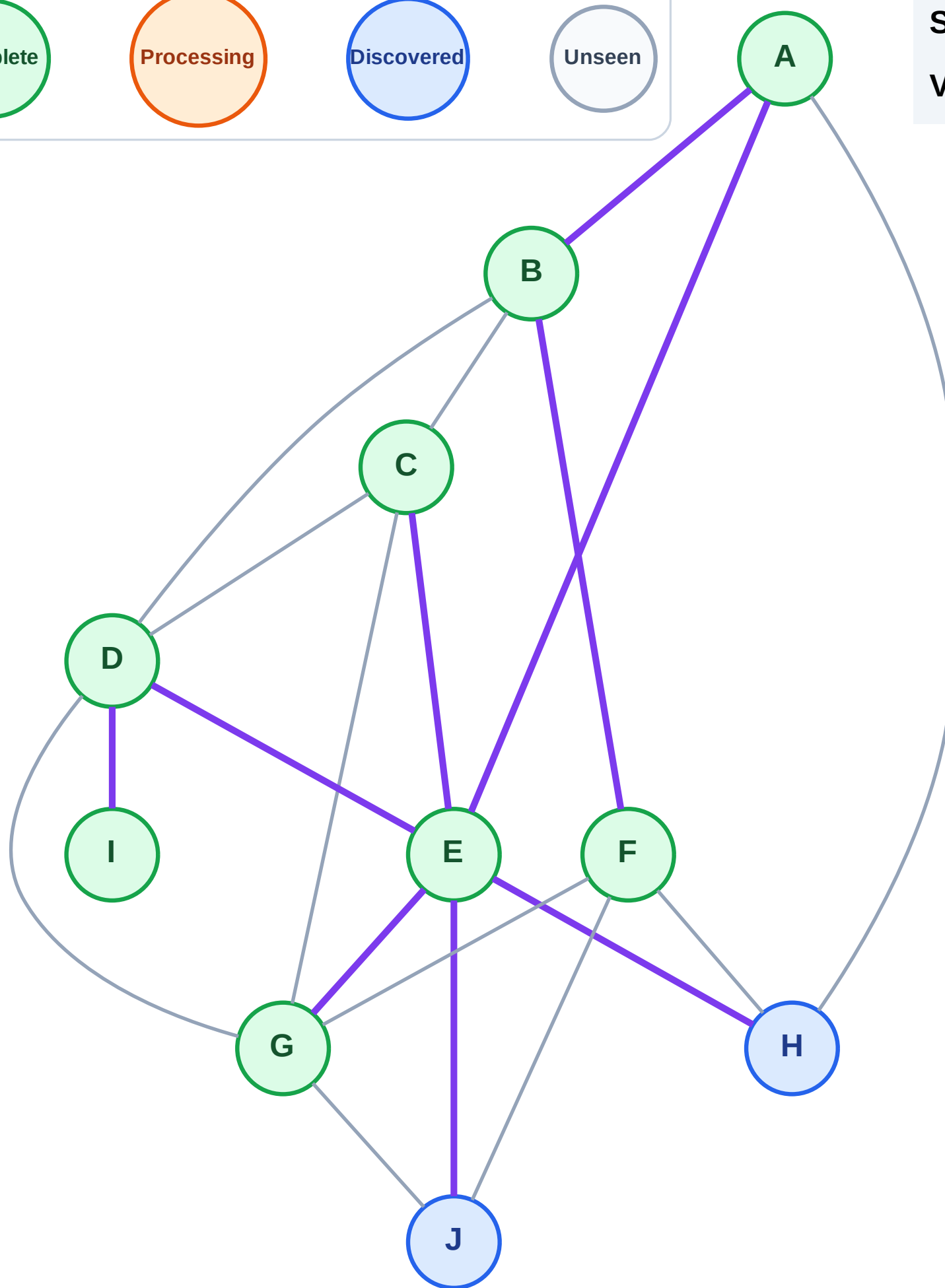
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

Complete

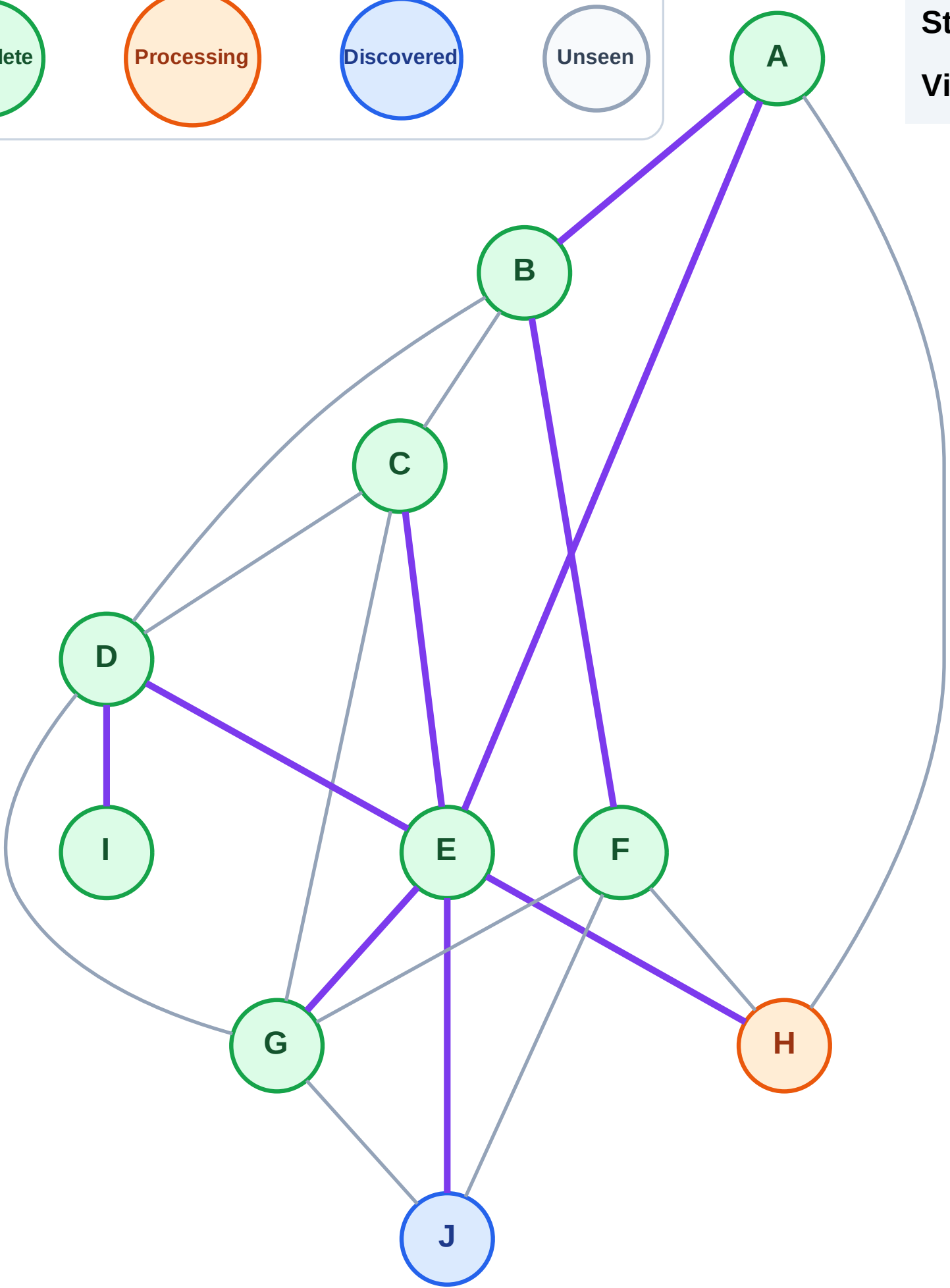
Processing

Discovered

Unseen

Stack (top at right): J

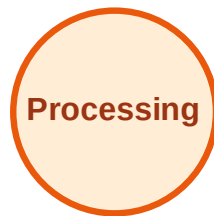
Visited: A, B, C, D, E, F, G, I



DFS Traversal

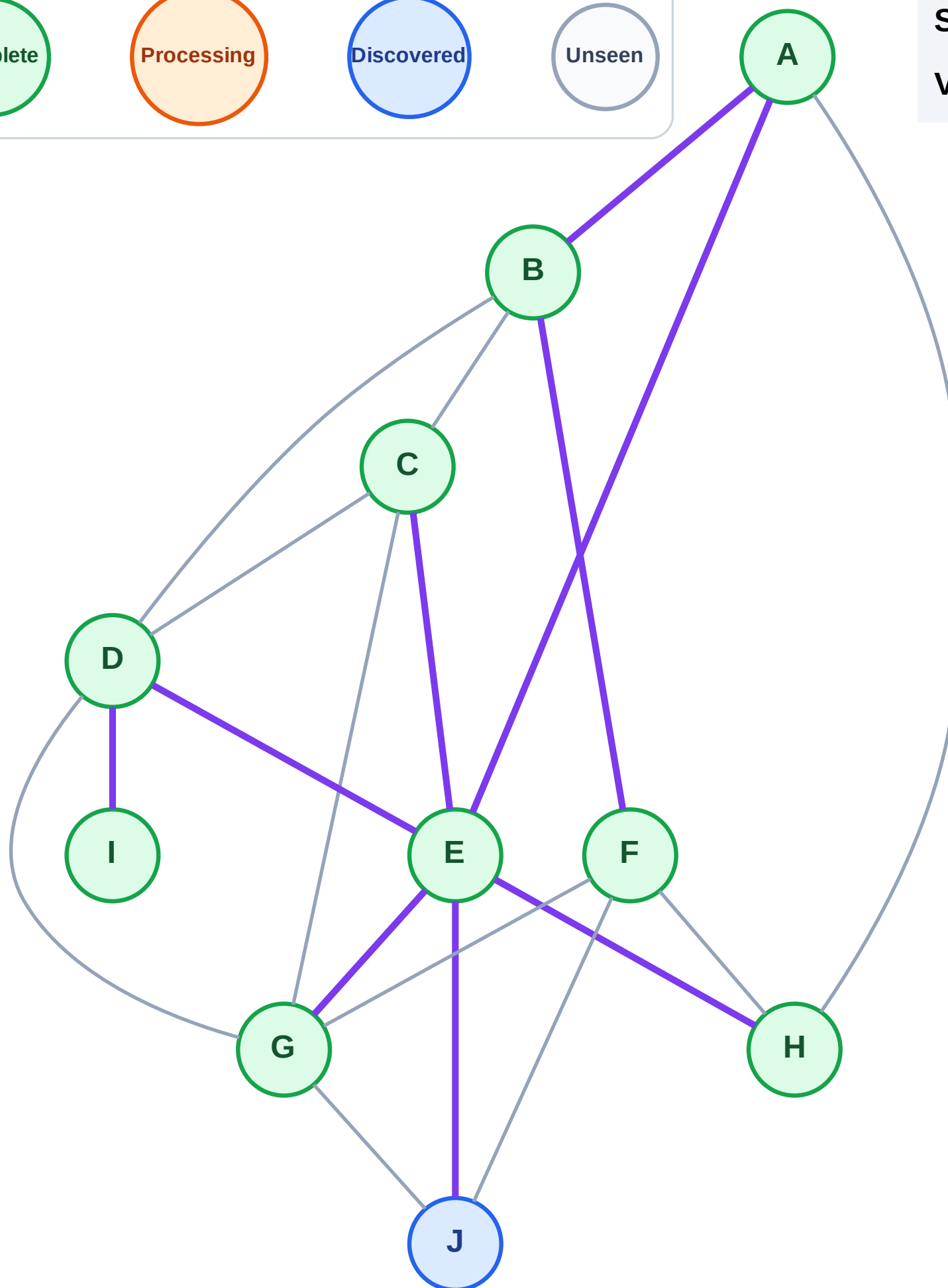
Step 19: No new neighbors from H

Legend



Stack (top at right): J

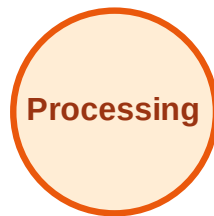
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

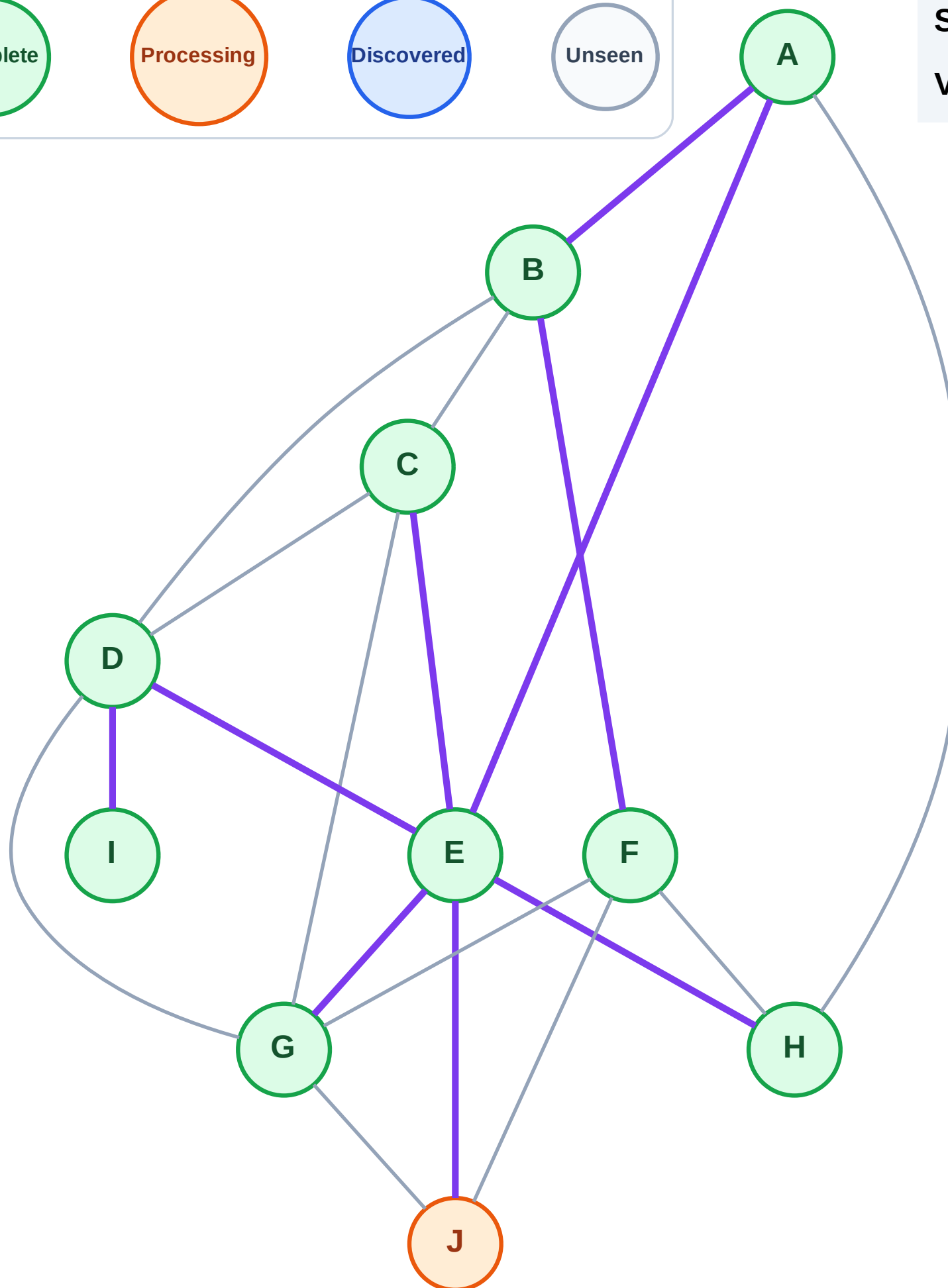
Step 20: Process node J

Legend



Stack (top at right): (empty)

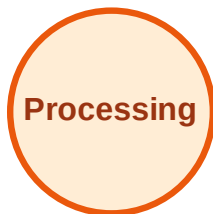
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

